The Cerebellum (2026) 25:15
https://doi.org/10.1007/s12311-025-01939-3

REVIEW

Consensus Paper: Models of Cerebellar Functions

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Received: 26 June 2025 / Accepted: 21 November 2025 / Published online: 9 February 2026

Abstract

For a long time, from the nineteenth century to most of the twentieth century, the cerebellum was thought to be an organ that regulates movement. Towards the end of the twentieth century, the brain functions associated with the cerebellum began to extend beyond motor control. Now, there is a consensus that the cerebellum is involved not only in motor functions but also in the most basic autonomic functions and the most complex cognitive and emotional functions, with a focus on predictions and internal models. A new functional model of the cerebellum is needed to explain all layers of brain functions by extending predictive computations in the cerebellum. On the other hand, the cerebellum and the basal ganglia were believed to be independent and complementary motor centers that lacked direct neural connections. For example, in neurophysiology classes in the 1980s, the characteristics of cerebellar ataxia were summarized as hyperkinetic and hypotonia, while the characteristics of Parkinson's disease (traditionally classified as “basal ganglia disorder”) were summarized as hypokinetic and hypertonia, and therefore their functions were assumed at opposite poles, without interactions between the two main subcortical systems. The cerebellum and the basal ganglia were also assigned contrasting models regarding their learning mechanisms. Namely, the cerebellum was assumed to employ supervised learning with error signals, while the basal ganglia were assumed to employ reinforcement learning with reward prediction errors. However, recent neuroanatomical studies have demonstrated a number of novel connections between them, questioning their independence. Moreover, recent single-neuron recording and inactivation studies provided evidence that the cerebellum may also be involved in reinforcement learning. The cerebellum is neither independent of the basal ganglia nor exclusively specialized for supervised learning. We now need a new, general model to explain the contradiction between the known uniformity of the cerebellar cortex's structure and the newly added diversity of brain functions to which the cerebellum contributes. This consensus paper summarizes many of the seeds of such a new theory. The panel of experts (1) highlights the importance of the anatomical connectivity between cerebellar circuitry and basal ganglia, (2) points out that the anatomy of the cerebellum is unique and allows predictive computations in motor and extra-motor domains such as cognition, affect, social interactions and reward processes, (3) underlines the need to further elucidate the nature of interactions between cerebellar cortex and cerebellar nuclei to better understand cerebellar and psychiatric disorders and (4) suggests that common operations may underlie the motor and non-motor functions of the cerebellar circuitry. Cerebellar models remain a major topic of research to improve our understanding of the numerous cerebellar activities and to better understand the complexity of cerebellar disorders.

Keywords Ataxias · Cerebellum · Prediction · Forward Model · Kalman Filter · The Cerebellar Cognitive-Affective Syndrome

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Abbreviations

CA	Cerebellar ataxia
CF	Climbing fiber
CS	Complex spike
CNMC	Corticonuclear microcomplex
DC	Dentate cell
DCN	Deep cerebellar nuclei
DN	The dentate nucleus
DoT	Dysmetria of thought
GC	Granule cell
GPe	The external segment of the globus pallidus
GPi	The internal segment of the globus pallidus
IMCA	Immune-mediated cerebellar ataxia
IO	Inferior olive
LTD	Long-term depression
LTP	Long-term potentiation
M1	The primary motor cortex
MAI	Marr-Albus-Ito
MF	Mossy fiber
MLI	Molecular layer interneuron
MOSAIC	Modular selection and identification for control
MRI	Magnetic resonance imaging
NHP	Non-human primate
NICS	Non-invasive cerebellar stimulation
PC	Purkinje cell
PF	Parallel fiber
PM	Premotor cortex
PN	Pontine nuclei
PO	Primary olive
RNN	Recurrent neural network
RNp	The parvocellular part of the red nucleus
rPE	Reward prediction error
SCA	Spinocerebellar ataxia
SNpc	The substantia nigra pars compacta
SNpr	The substantial nigra pars reticulata
SS	Simple spike
STN	The subthalamic nucleus
TD	Temporal-difference
TMS	Transcranial magnetic stimulation
UCT	Universal cerebellar transform
VL	Ventrolateral
VNC	Vestibular nucleus cell
VOR	Vestibulo-ocular reflex
VTA	Ventral tegmental area

Introduction (Shinji Kakei)

The history of cerebellar research has been characterized by several milestones. In the early twentieth century, cerebellar functions were hypothesized as lost functions of patients with cerebellar ataxia (CA)[1, 2]. The most recognized deficits included *deteriorated rhythmicity* of alternate movements (*adiadochokinesis*), *lack of coordination* in reaching movements leading to overshoot/undershoot (*dysmetria*), and rhythmic involuntary movements (*tremors*). There was a consensus that the cerebellum is critical in coordinating movement by organizing the timing of muscle activities. This interpretation was unsatisfactory because it did not include a cerebellar mechanism for coordination.

We had to wait for the seminal monograph by Eccles, Ito, and Szentágothai [3] before outlining the dynamics of the cerebellar machinery. They transformed our understanding of the cerebellum by visualizing the information flow in the cerebellar circuitry. Nevertheless, there remained a gap between the cerebellar circuit dynamics and the behavioral phenotypes of CA. Therefore, they disclosed their findings and invited new interpretations.

Their efforts were rewarded soon. The earliest theories were the learning theories of Purkinje cells (PCs) by Marr (1969)[4] and Albus (1971)[5]. They noted the unique arrangement of the climbing fiber (CF) and parallel fiber (PF) inputs to a PC. They proposed learning models of a single PC in which the sole CF input provides a learning signal to modulate synaptic efficacy for PF inputs to the PC. Their theories raised two critical questions: 1) the mechanism to select or unselect a specific PF synapse for a gain change; 2) what is encoded in the CF activities. Their cerebellar models were mostly confined to the cerebellar cortex and almost excluded the deep cerebellar nuclei (DCN), the major output of cerebellar circuitry, and the extra-cerebellar targets.

Masao Ito, one of the authors of the monograph, developed models that integrated the cerebellar cortex, the DCN, and the extra-cerebellar targets (1970)[6]. He modeled different cerebellar regions by introducing distinct schemes of control engineering. First, he modeled the vestibulo-ocular reflex (VOR) by the Flocculus as a *feedforward control system* (Fig. 6C in [6]). The VOR stabilizes retinal images by rotating the eyes in the opposite direction of the head motion. Note that the direction, amplitude, and gain of the reflex are prepared *predictively* in the flocculus, depending on the head's motion. Next, he modeled the *newer part* of the cerebellum, the cerebrocerebellum (neocerebellum), to explain the control of voluntary movement. He proposed that the cerebrocerebellar loop is modeled as a *model reference adaptive control system* (Fig. 7B in [6]). In modern terms, this model is identical to the *internal forward model*, which predicts future states.

Since then, we have seen a massive expansion of the cerebellar functional domains. Leiner and colleagues (1986) [7] raised a question about the long-standing belief that the cerebellum was an organ for motor coordination. They suggested cerebellar contribution to cognitive or verbal functions [7, 8]. Schmahmann reviewed historical evidence pointing to cerebellar roles beyond motor control and introduced the dysmetria of thought theory [9], and together with Deepak Pandya described feed forward projections from cerebral association and paralimbic cortices to the pons in monkey [10–15] which were complemented by the evidence of feedback projections from cerebellar outputs to the prefrontal cortex by Strick and colleagues [16, 17]. Cerebellar researchers were initially skeptical about the new idea because at that time evaluation of *non-motor* functions relied on the *manipulation* of devices, Ito was among the earliest advocates of the cerebellar contribution to higher brain functions [18], adapting his *internal forward model* [6] designed for motor control to the *control* of *mental* (verbal or nonverbal) and *autonomic* functions. Eventually, the two issues were resolved [16, 19–24], and the cerebellar contribution to higher brain functions was confirmed, as exemplified by the description of "the cerebellar cognitive, affective syndrome" [24].

The expansion of the cerebellar territories is continuing. For instance, the cerebellum and the basal ganglia were thought to form *separate* loops with the cerebral cortex: the cerebro-cerebellar loop and the cerebro-basal ganglia loop [25]. The reason for this belief was that the projections from the cerebellum and the basal ganglia have little overlap in the thalamus [25]. In this view, the cerebral cortex was pivotal in integrating inputs from the cerebellum and the basal ganglia. However, transneuronal tracing studies by Strick and his group [26, 27] revealed more *direct* connections between the cerebellum and the basal ganglia. Indeed, recent physiologic studies revealed the reward-expecting activities in granule cells (GCs)[28, 29], which was a hallmark of the nigral dopaminergic neurons.

In summary, the cerebellum may also engage in reinforcement learning with reward signals, in addition to supervised learning with error signals [30]. Finally, we now know that the cerebellum is connected to the limbic system and the autonomic nervous system [31–33]. These findings indicate a need for a major update of cerebellar models proposed for motor control (see also a review by D'Angelo and Casali [34] on this issue).

The current Consensus Paper gathers an international panel of experts working on models of cerebellar functions from the physiological, morphological, theoretical, and clinical points of view. This article contains, in addition to the Introduction and Discussion, fifteen sections that deal with the expansion of the cerebellar influence in the brain, updates on the Marr-Albus-Ito model, updates on the cerebellar learning, updates on the cerebellar internal models, and more. The panel of experts proposes seeds for novel models of cerebellar function on the basis of advances of the last decades.

Connections between the Cerebellum and the Basal Ganglia (Andreea C. Bostan and Peter L. Strick)

The organization of cerebellar connections with the cerebral cortex provides an anatomical framework that has closely shaped conceptual models of cerebellar function. The cerebellum receives inputs from broad areas of the cerebral cortex via the pontine nuclei (PN)[35]. Traditionally, the cerebellum was believed to integrate signals from the cerebral cortex and influence only a single cerebral cortical area – the primary motor cortex (M1) – through projections to the ventrolateral (VL) thalamus [36]. Within this traditional framework, the cerebellum was thought to play an exclusive role in motor function.

More recent findings have revealed a more complex perspective on cerebellar functions. The cerebellum projects not only to the VL thalamus, but to multiple other thalamic nuclei [37]. These broader projections enable the cerebellum to communicate with many of the cerebral cortical areas that provide it with inputs, thereby influencing a variety of motor and non-motor functions [38, 39]. Further, studies using trans-synaptic tracers in non-human primates (NHPs) have demonstrated that cerebellar connections with the cerebral cortex are topographically organized in the form of close-loop circuits: cerebellar regions that receive inputs from specific cortical area also send outputs back to the same cortical regions [20]. These findings helped define separate motor and non-motor domains within the DCN [40] and the cerebellar cortex [20]. The non-motor domains are extensive, suggesting that a significant portion of the cerebellum is dedicated to functions beyond motor control [38, 39, 41–43]. Cerebellar contributions to non-motor functions are further discussed in Chapters 2 and 13.

Like the cerebellum, basal ganglia connections with the cerebral cortex are topographically organized into parallel closed-loop circuits supporting motor and non-motor functions [44]. Although both subcortical structures influence many of the same cerebral cortical areas, they do so through distinct thalamic nuclei [37, 45], with minimal direct overlap [46]. This anatomical separation at the level of the thalamus has supported the view that the cerebellum and the basal ganglia have largely independent functions, with any meaningful interactions occurring at the level of the cerebral cortex [47].

More recent results from neuroanatomical tracing studies in NHPs have identified pathways that allow for more direct communication between the cerebellum and the basal ganglia, challenging the notion of their functional independence [26, 48]. These findings have contributed to a growing perspective that the cerebellum, basal ganglia, and cerebral cortex form an integrated network [27]. Here, we provide a brief overview of these findings (Fig. 1) and highlight areas where further investigations are needed.

Studies in NHPs have used the retrograde transneuronal transport of rabies virus to reveal pathways linking the cerebellum and basal ganglia. Rabies virus is transported across synapses exclusively in the retrograde direction in a time-dependent manner, allowing for the tracing of multi-synaptic circuits [49]. Rabies virus injections into the striatum revealed di-synaptic projections from the DCN, particularly the dentate nucleus (DN) [26]. This pathway is mediated through the intralaminar thalamic nuclei [26, 50]. Studies in rodents indicate that through this pathway, the cerebellum can drive activity of striatal neurons and change aspects of cortico-striatal plasticity [51].

With longer survival times allowing for transport across three synapses, injections of rabies virus into the external segment of the globus pallidus (GPe) also resulted in rabies-virus neurons labeled in the dentate, while injections into the internal segment of the globus pallidus (GPi) did not [26]. These results suggest that cerebellar output may preferentially influence basal ganglia pathways through the GPe, a key relay in the "indirect pathway," as opposed to the "direct pathway" through the GPi. Although functional studies of the pathway from the cerebellum to the basal ganglia in rodents [51] can also be interpreted to suggest a bias towards the indirect pathway [27], further studies are needed to examine the extent and degree of specificity in cerebellar projections to the basal ganglia. Notably, cerebellar projections to the GPe appear to be topographically organized, with distinct GPe regions receiving inputs from different areas of the dentate [26]. Further, both motor and non-motor regions of the dentate send projections to the GPe [26], indicating that cerebellar outputs are likely to contribute to both motor and non-motor functions of the basal ganglia.

Relatively few studies have examined reciprocal pathways through which the basal ganglia may modulate cerebellar function. Early electrophysiological studies in cats [52–55] suggested the existence of such pathways, but a specific anatomical connection was revealed only recently. A rabies virus tracing study in NHPs revealed a substantial di-synaptic pathway from the subthalamic nucleus (STN) to the posterior lateral cerebellar cortex [48]. In this study, injections into a motor region of cerebellar cortex (lobule

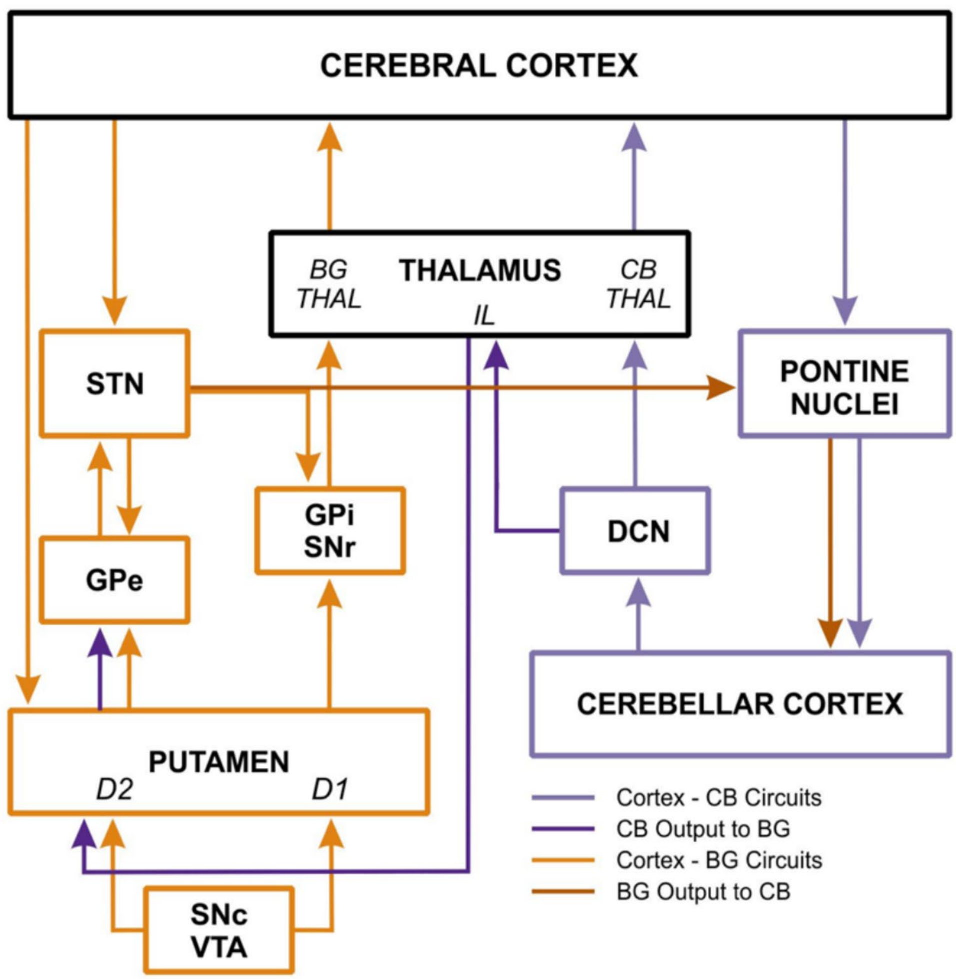


Fig. 1 Schematic diagram of the circuits that link 1) the cerebellum (CB) with the cerebral cortex (purple arrows) and 2) the basal ganglia (BG) with the cerebral cortex (orange arrows). Cerebellar and basal ganglia outputs to the cerebral cortex are mediated through distinct regions of the thalamus (THAL). Cerebellar outputs to the basal ganglia (dark purple arrows) are mediated through the intralaminar nuclei (IL) of the thalamus. Basal ganglia outputs to the cerebellum originate from the subthalamic nucleus (STN) and are likely mediated by the pontine nuclei (dark orange arrows). BG, basal ganglia; CB, cerebellum; D1 and D2, medium spiny neurons expressing dopamine receptors 1 and 2; DCN, deep cerebellar nuclei; GPe and GPi, external and internal segments of the globus pallidus; IL, intralaminar thalamic nuclei; SNc and SNr, pars compacta and pars reticulata of the substantia nigra; VTA, ventral tegmental area

HVIIB) labeled neurons in motor portions of the STN, while injections into a non-motor region (Crus II) labeled neurons in associative regions of STN, indicating a topographically organized connection. Notably, no evidence was found for di-synaptic input to the lateral cerebellar cortex from the main output nuclei of the basal ganglia, the GPi and SNpr.

A study using the rabies virus tracing in rats confirmed the presence of a similar pathway in rodents [56]. However, unlike in primates where the STN projects to the lateral cerebellum, the rodent STN primarily targets the cerebellar vermis and lacks apparent topographic organization. Despite consistent physiological evidence supporting the relevance of this pathway in rodents [57–59], its prominence and function may differ from primates. More recently, alternative pathways through the pedunculopontine tegmental nucleus [60], the zona incerta [61], or the inferior olive (IO) [62] have been proposed as potential routes for basal ganglia influence of the cerebellum in rodents. These pathways remain to be functionally characterized in rodents and validated in primates.

Several interesting lines of research point to additional interactions between the cerebellum and the basal ganglia that are mediated through the dopaminergic midbrain. Early anatomical and electrophysiological studies in cats [52, 63], along with modern viral-genetic circuit tracing and optogenetic approaches in mice [64–66], suggest that the cerebellum can modulate basal ganglia activity via the dopaminergic midbrain. Cerebellar projections appear to target both dopaminergic and non-dopaminergic neurons in the ventral tegmental area (VTA) and the substantia nigra pars compacta (SNpc), allowing the cerebellum to influence dopamine levels in the basal ganglia during movement and reward processing [67]. Tractography studies in humans further support the presence of cerebellar projections to the dopaminergic midbrain [68, 69]. In turn, dopaminergic and non-dopaminergic neurons in the rat VTA send projections to both the cerebellar cortex (primarily the hemisphere) and the deep cerebellar nuclei (primarily the lateral nucleus) [70, 71]. Recently, studies in the mouse have begun exploring functional roles of dopaminergic receptors in the cerebellar cortex [72] and the deep cerebellar nuclei [73, 74]. However, it is worth nothing that although catecholaminergic and dopaminergic innervation has been described in primates [75], the distribution of catecholaminergic afferents in the cerebellum is variable across species, indicating that there may be species differences in the functional role of dopaminergic influence over the cerebellum [76]. Overall, pathways linking the cerebellum with the dopaminergic midbrain remain to be characterized in primates, through anatomical and functional studies.

While further investigation of the individual pathways is needed, the discovery of anatomical connections between the cerebellum and the basal ganglia reinforces the idea that these structures form a densely interconnected network, both with each other and with the cerebral cortex [27]. Importantly, these findings are reshaping traditional concepts of cerebellar function by prompting new questions about its role in processes that have been historically attributed to the basal ganglia. This expanding line of research highlights the cerebellum's involvement in reward and emotional processing [77–80] (see also Chapter 8) and underscores its clinical relevance to a broad spectrum of neurological and psychiatric conditions (see also Chapter 14), including Parkinson's disease [81, 82], dystonia [83, 84], Tourette syndrome [85–87] and addiction [88].

Cerebellar Contributions to Cognitive Functions (Xavier Guell and Jeremy D Schmahmann)

Dysmetria of Thought (DoT) [9, 89, 90] is an overarching conceptual model of the cerebellar contribution to cognition, emotion, and motor control. The model holds that the cerebellum regulates the speed, capacity, consistency, and appropriateness of mental or cognitive processes in the same way that it regulates the rate, force, rhythm, and accuracy of movements (see also Introduction).

How is this possible, and what underlying mechanism explains such a model? DoT is predicated on the theory of a Universal Cerebellar Transform (UCT) [91], which arises from the complimentary features of essentially homogenous intrinsic cerebellar cortical architecture and repeating corticonuclear microcomplexes (see also Chapters 10 and 15), set against heterogenous cerebellar connections with extracerebellar structures (see also Chapter 15).

Cytoarchitectonic variations, Brodmann areas that define cerebral cortex, are absent in the cerebellar cortex, which instead is arranged in a highly regular, paracrystalline manner [92]. Minor variations exist, as in zebrin bands with different physiological parameters [93] but the near uniformity of cerebellar cortical architecture is a defining feature of nervous system organization [94].

In contrast, functional heterogeneity in the cerebellum is evident in task-based and resting state connectivity functional MRI (magnetic resonance imaging), and connectional heterogeneity is documented in anatomical tract tracing studies (see also Chapter 15). Spinal cord, primary motor cortex and motor thalamus are linked with cerebellar cortical lobules I-VI and VIII [20] which activate in human functional MRI motor tasks [95, 96]. The two motor representations (lobules I-VI and lobule VIII) are followed by a motor-fugal gradient from lobules I-VI towards Crus I (first non-motor representation), from lobule VIII towards

VIIB and Crus II (second non-motor representation), and from lobule VIII towards IX and X (third non-motor representation) [97, 98] (Fig. 2A). The motor-fugal gradient of cerebellar cortical organization echoes that of the cerebral cortex [99], progressing from primary motor processing towards unimodal association functions, followed by multimodal association functions [97]. Similar functional heterogeneity and gradient ordering is observed in the cerebellar DN, as shown in monkey tract tracing investigations [17] and gradient-based functional imaging studies in humans [100] (Fig. 2B, C). The anatomical relationship of primary processing to unimodal and multimodal association processing is fundamentally different in cerebral cortex compared to cerebellum. In cerebral cortex, association fibers support propagation, synthesis, and abstraction of information – from primary motor activation, to motor planning, to abstract planning – as activity from one area influences the next. In cerebellar cortex and nuclei, there is an identical ordering of functions, but unlike cerebral cortex, the cerebellum has no association fibers linking one region of cortex with another. In cerebellar cortex and nuclei, therefore, it is not association fibers, but ordered connections to cerebral cortex via relays in thalamus [101] and ordered connections from cerebral cortex via relays in PN [10, 14, 89, 102] that dictate the position of and relationship between functional domains.

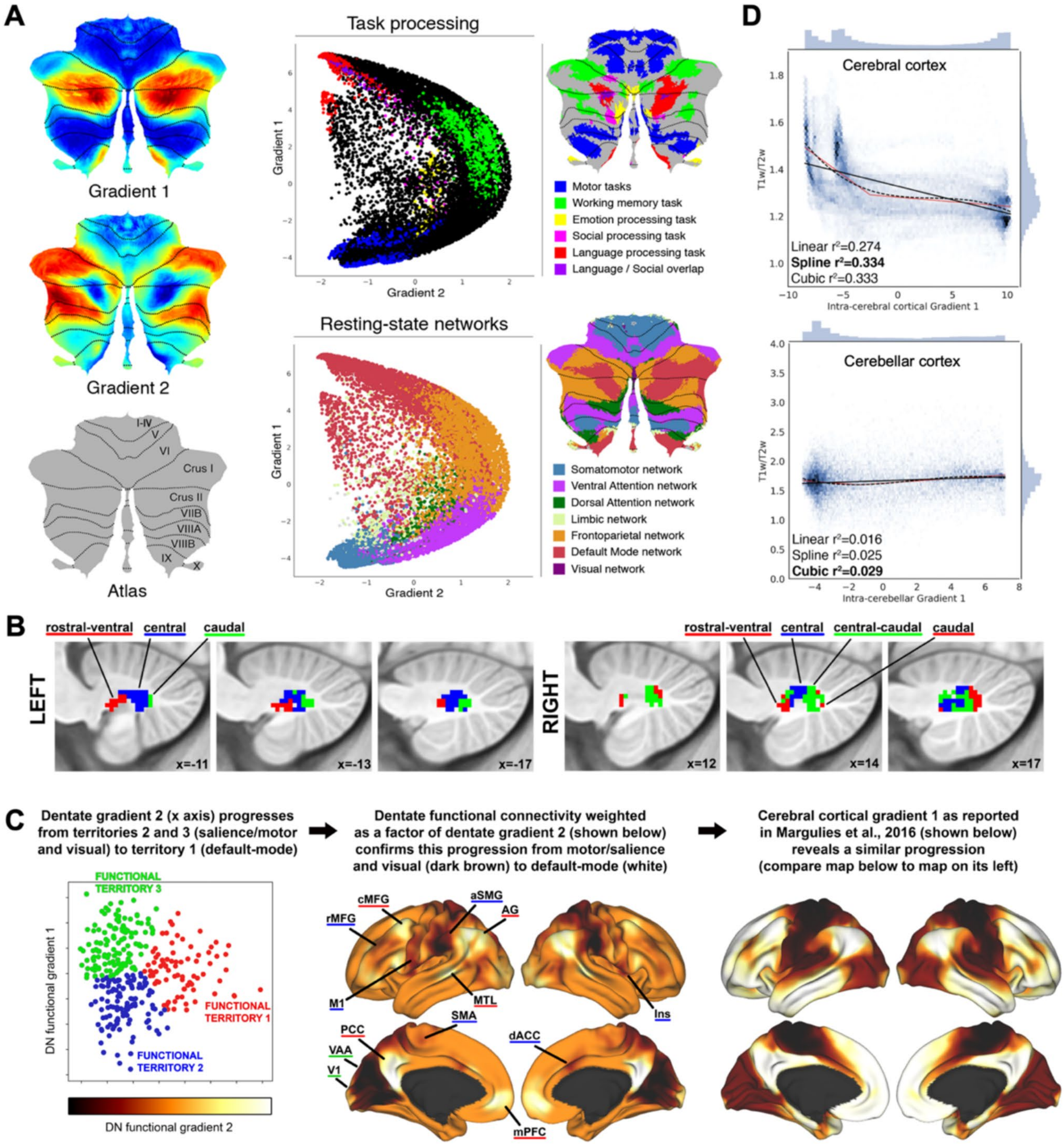
The DoT model is informed by these anatomical realities. In line with the Schmahmann and Pandya principles of cerebral cortical organization [103], the UCT theory holds that the unique architecture of the cerebellum determines a matching unique information transform. The behavioral outcome of this transform is that cerebellum modulates, rather than generates behavior. It maintains function around a homeostatic baseline, automatically and without conscious awareness, serving as an oscillation dampener to optimize performance according to context, facilitating actions harmonious with goals, and judged accurately and reliably according to strategies mapped prior to and during behavior. The cerebellum detects, prevents, and corrects mismatches between intended outcome and perceived outcome of the organism's interaction with the environment [9, 91, 94].

Through the heterogeneity of cerebellar connections with extra-cerebellar structures, the UCT is applied to diverse streams of information processing across multiple domains of sensorimotor, vestibular, autonomic, limbic, and cognitive behaviors (see also Chapter 15).

The corollary is that the cerebellum works the same way across domains in health, and breaks the same way across domains in disease. Anterior lobe lesions damaging the cerebellar motor representation produce dysmetria of movement, with ataxia, dysmetria, and dysarthria; cerebellar vestibular lesions produce the vestibular syndrome; and posterior lobe lesions of the cognitive-limbic cerebellum produce disorders of intellect and emotion, the cerebellar cognitive affective/Schmahmann syndrome, the third cornerstone of clinical ataxiology. [104, 105]. The behavioral analogy is that just as disrupted cerebellar motor behavior does not lack power but fails in precision and coordination, cerebellar cognitive manifestations do not include amnestic dementia, but disrupt mental control and context-appropriate interaction. Mental processes are imperfectly conceived, erratically monitored, and poorly performed. There is an unpredictability to social and societal interaction, a mismatch between reality and perceived reality, and erratic attempts to correct the errors of thought or behavior. In language, for example, isolated cerebellar injury does not usually result in aphasia, but rather, language production may be impaired and lack context-appropriateness, and comprehension may fail to adjust to figurative or ambiguous linguistic elements [106].

The DoT model provides testable hypotheses. It predicts that topographic distribution of functional domains in cerebellar cortex is determined by anatomical connectivity to extracerebellar nodes, not by variations in cerebellar cortical microstructure. We tested this using T1w/T2w MRI intensity as a proxy measure of microstructure, and functional gradients derived from resting-state connectivity as a proxy measure of functional specialization. As predicted by the model we showed a correlation between the two in

Fig. 2 Imaging analyses of the human cerebellar system. (A) Cerebellum gradients (from [97]) and relationship with discrete task activity maps (from 96]) and resting-state maps (from [95]). Gradient 1 extended from default mode network to motor regions. Gradient 2 isolated working memory/frontoparietal network areas. In the scatterplots, each dot corresponds to a cerebellar voxel, position of each dot along x and y axis corresponds to position along Gradient 1 and Gradient 2 for that cerebellar voxel, and color of the dot corresponds to task activity (top) or resting-state network (bottom) associated with that particular voxel. Note that the language task subtracts a "story listening" condition minus a "doing math" condition, isolating language but also subtracting task-focused processing, therefore resulting in a map similar to the default-mode network. (B) Spatial location of functional territories in human DN. Territories in red mapped to default-mode network areas of the cerebral cortex. Territories in blue mapped to motor and salience network areas. Territories in green mapped to visual processing areas, including visual association processing areas. Adapted from [100]. (C) The second functional gradient of human DN, when projected to the cerebral cortex, replicates the principal gradient of cerebral cortical organization that progresses from primary processing to default-mode network areas. Adapted from [100]. (D) Using T1w/T2w MRI signal as a proxy for cerebral cortical and cerebellar cortical microstructural organization, and functional gradients as a proxy for functional specialization, preliminary evidence shows that functional specialization is independent of microstructural variation in cerebellum (right column, note lack of correlation between the two axes) but not in cerebral cortex (left column). Studies with higher MRI resolution will be needed to confirm this observation. Adapted from [107]



the cerebral cortex, but not in the cerebellar cortex [107] (Fig. 2D).

DoT does not contradict other models of cerebellar function described in this article, but it is unique in its use of phenomenological resemblance as a tool for neurological analysis. The DoT model stems from the clinical observation that patients with cerebellar injury demonstrate abnormalities in thought and emotion that resemble their abnormal motor behavior. This use of phenomenological resemblance, applied to patients with cerebellar injury, forms the basis of DoT. The same use of phenomenological resemblance, applied across all areas of neurology, is the basis of the newly proposed concept of *Phenoconsonance*. Phenoconsonance is inspired by DoT, provides broad clinical and anatomical context for the ideas discussed in this section, and proposes new routes for research in clinical-anatomical relations (see also [108]).

Other recent developments in this area include lesion-behavioral data in rhesus monkeys in which bilateral lesions of ventral sectors of the dentate nuclei impair tasks of working memory (Delayed Recognition Span Task [109]) and executive function (Conceptual Set Shifting Task [110]) but spare tasks of manual dexterity (Kuypers' Task [111]) and recognition memory (Delayed Non-Matching to Sample task [112, 113]). This study provided the first empirical support for anatomical and functional MRI evidence of a motor-cognitive dichotomy in the cerebellar dentate nucleus. It also identified a within-cognition dichotomy in which cerebellum modulates dorsal stream cognitive functions (spatial awareness; dynamic actions of where and how) characterized by executive functions including working memory, but is not engaged in ventral stream cognitive processes (static actions of object identification) exemplified by recognition memory [113]. This is in line with anatomical data showing that the pontine relations of the two visual streams are different [12]. Whereas the dorsal stream connections are prominent, the ventral stream has few or no pontine connections, a finding that led to the prediction that within the visual realm, the corticopontocerebellar system is more concerned with visual spatial related functions and parameters of visual motion than with functions related to visual object identification or discrimination [12]. These anatomical and lesion-deficit behavioral studies confirm and extend our interpretation of the Dysmetria of Thought theory that the nature of cerebellar contribution to behavior (how the cerebellum modulates information) influences the functional domains that the cerebellum is most relevant in (what information the cerebellum modulates). The theoretical complexities of the distinction between how function is modulated, and what function is modulated, are further expanded in [108].

The Relationship between the Marr-Albus and the Ito Models (Hirokazu Tanaka)

The Marr-Albus-Ito (MAI) model, proposed in the late 1960 s and early 1970 s, remains the most influential computational account of cerebellar functions [114, 115]. Importantly, the MAI framework is not a single unified model but rather the conjunction of two conceptually distinct complementary approaches: The Marr-Albus perceptron model [4, 5] and the Ito internal model [6]. We argue that the perceptron model of the cerebellar cortex proposed by Marr and Albus and the internal model of the cerebellum proposed by Ito are best understood in the light of computational levels. In theoretical neuroscience, computational models are classified asinterpretive, descriptive, or mechanical [116]. First, an interpretive model formulates the goal of neural computation (what objective the brain must solve for) and explains behavioral and neurophysiological findings by optimizing specific objective functions. This class of models corresponds to Marr's level of computational theory and includes the information-maximization model of sensory processing [117, 118] and the optimal feedback model for motor control [119]. Second, a descriptive model characterizes a concise statistical relationship between neural activities and individual behaviors, revealing what behaviors the neural activities represent. This class of models discusses the brain's representations for neural computation, thereby matching with Marr's level of representations and algorithms. Examples of descriptive models include the receptive-field model of the primary visual cortex [120] and the population-vector decoding algorithm of upper-limb movements [121]. Finally, a mechanical model posits how a biological mechanism implements specific neural information processing, comparable to Marr's level of physical implementations. Mechanical models include the Hodgkin-Huxley model of generating action potentials in a single neuron [122] and the dopamine-neuron implementation of temporal-difference (TD) errors in reinforcement learning [123].

The Marr-Albus perceptron model of the cerebellum describes how a single PC performs pattern classification and learns from experiences, thereby falling into the mechanistic model class [4, 5]. Inspired by the uniform anatomical structure in the cerebellar cortex, Marr and Albus independently proposed the perceptron learning model of PCs; a PC performs pattern classification of input from the PF pathway and modifies the PF synapses when the CF pathway conveys a supervising signal (see also Chapter 5). The GCs expand the input patterns from the MFs, simplifying the classification problem of PCs. The perceptron model has been instrumental in both experimental and neurophysiological studies. The key prediction of the perceptron model was the synaptic plasticity of PF synapses following a CF input. Experimentally stimulating the CF of a PC induced long-term depression (LTD) of PF synapses in the same neuron, confirming the model's prediction [124]. Theoretically, the original perceptron model computes the pattern classification of static patterns and is extended to handle dynamic patterns, such as the adaptive filter model [125] and the liquid-state-machine model [126]. In summary, the Marr-Albus perceptron model and its extensions describe how the cerebellum learns static and dynamic patterns from supervising signals conveyed from the CF. However, the Marr-Albus perceptron model did not clarify what the cerebellum learns.

The Ito internal model specifies what the cerebellum computes in motor control and cognitive processing, therefore classified as an interpretative model [6]. At the same time as Marr-Albus' perceptron model, Ito postulated an internal model of the cerebellum, stating that "the large loop through the external world may be effectively replaced by an internal one through the cerebellum which would serve as a model of the combination of the spinal motor system, the external world and the sensory pathways." Therefore, the internal model defines the goal of cerebellar computation as learning the dynamics of the body, the peripheral nervous system, and the external world. There are two types of internal models: (i) a forward model, which predicts future state from a current estimate and an efference copy (see also Chapters 7,9,11, and 15), and an inverse model, which computes control signals to achieve a desired state (see also Chapter 12). In modern terminology, the internal model Ito proposed is best regarded as an internal forward model. There is accumulating evidence that the cerebellum functions as an internal forward model in behavioral, neuroimaging, and neurophysiological studies [127]. Our computational analyses suggest that PCs in the cerebellar cortex perform predictive computation of motor consequences and that cerebellar nucleus cells integrate these PC predictions with sensory feedback in an optimal way (see also Chapter 10). The same cerebellar circuit computation is also supported in a recent review (see Fig. 2b in [128]). Initially, the internal model was applied mainly to motor control and motor learning, but recently, it has been extended to cognitive and affective information processing. Therefore, the cerebellum predicts the consequences of motor, cognitive, and social actions [129]. We conclude that the Marr-Albus and the Ito models are complementary; the cerebellum predicts and learns the dynamics of our body, external world, and other minds (the Ito internal model), and the PCs subserve the predictive computation by modifying the strength of PF synapses according to the supervising signals from CFs (the Marr-Albus perceptron model).

To Time or to Tame, is that still a Question? (Tadashi Yamazaki)

A theory is a way of thinking about what a given system is and how it behaves. A good theory has four important properties: (1) it is grounded in available experimental data, (2) it provides experimentally testable predictions, (3) it is consistent with data obtained in the future, and (4) it encourages the emergence of more advanced theories.

Around the mid-1960s, following the discovery of CF activation of PCs [130, 131], questions were raised regarding the functional significance of this powerful synaptic input. In the next few years, two theories were proposed following the discovery. Rodolfo Llinás proposed that the CF plays a role in rapid (phasic) motor timing [132, 133], whereas Masao Ito stated that the CF provides instruction signals for motor learning [6, 134, 135]. Since then, timing vs learning became a long-lasting issue regarding the role of CF in the cerebellar research. However, in 1960-70 s, any synaptic plasticity that could underline motor learning was not known on cerebellar PC dendrites.

In 1982, Ito and his colleagues discovered that conjunctive MF/PF and CF activation induces LTD at PF-PC synapses [136, 137]. Soon, Ito summarized the potential reasons why many earlier attempts to discover LTD were unsuccessful [138]. Moreover, in the same book, Ito addressed eight potential functional roles of CF besides the motor-learning mechanism, and argued that motor learning remained the most promising.

Meanwhile, the last review paper on the timing theory is written by the originator himself [139]. In that paper, two papers that seemed inconsistent with the learning theory were cited [140, 141], where intraperitoneal administration of T-588, a pharmacological inhibitor of LTD in vitro, did not impair motor learning in rats [140] and mice [141]. Schonewille et al. (2011)[141] also developed genetically engineered mice that lacked LTD induction, yet motor learning —assessed by VOR adaptation and delay eyeblink conditioning — was intact.

Then, once again, Ito summarized the potential reasons why those studies were inconsistent with the learning theory [142]. Following the Ito's summary, Anzai and Nagao (2014)[143] demonstrated that intraperitoneal T-588 application impaired motor learning tested by VOR adaptation in marmosets. Thus, compensatory mechanisms could modify synaptic plasticity in the remaining circuits for rodents in Welsh et al. (2005)[140] and Schonewille et al. (2011)[141]. A small note on Welsh et al. (2005)[140] is that the eyeblink conditioning paradigm that those authors used in the paper is categorized to trace but not delay eyeblink conditioning (0-trace conditioning), which is intact in LTD-deficient mice [144]. Finally, Yamaguchi et al. (2016)[145] confirmed normal LTD induction in the same genetically engineered mice [141] under certain experimental conditions. These findings strongly support the learning theory.

Furthermore, recent studies on information representation by CF inputs reveal non-motor aspects of CSs. In particular, CSs carry information on rewards and/or reward predictions [e.g., 28,146,147,148]. These findings fit well with the learning theory, whereas timing theory cannot account for them, because the timing theory is applicable to only motor functions.

These experimental findings stimulate theorists to advance the learning theory. While the original learning theory adopts supervised learning scheme (see also Chapters 5 and 7), recent theoretical studies propose reinforcement learning as the learning principle in the cerebellum [115, 149, 150] (see also Chapters 6 and 8).

In summary, the learning theory satisfies all the four properties described above. Ito's theory will live forever to open new eras in the cerebellar research.

As a final note, in Chapter 15 of Eccles, Ito, Szentágothai (1967)[3], there are two sentences that foresee the synaptic plasticity on PC dendrites based on the conjunctive activation of mossy and climbing fibers: "… it is important to examine the spatial and temporal relationships of actions produced by mossy and climbing fiber inputs that stem from some sensory input that is sharply localized in space and time" (p308), and "… that usage gives growth of the spines and particularly the formation of the secondary spines that Hamori and Szentágothai (1964)[151] described on Purkinje dendrites" (p314). This is remarkable, because this prediction was made 15 years before the discovery of LTD, and even a few years before the publication of Marr (1969)[4].

Rubral and Cerebral Origins of CF Error Signals in Reaching Adaptation (Shigeru Kitazawa and Masato Inoue)

Reaching, a fundamental voluntary movement, is adjusted to minimize end-point error, which becomes apparent when the visual field is displaced by wedge prisms. When a prism is introduced, the movement initially deviates in the direction of displacement, but this error decreases exponentially with each trial. Upon removal of the prism, reaching error in the opposite direction (after-effect), which again subsides exponentially (prism adaptation) [152]. Notably, this adaptation transfers minimally to the other arm when speeded reaching is required in both humans [153] and monkeys [154, 155], indicating that adaptation occurs in an arm-specific motor domain. Cerebellar deficits in humans, such as spinocerebellar ataxia (SCA) [156], infarcts in the distribution of the posterior inferior cerebellar artery [157], or cerebellar lesions in macaque monkeys [154], impair prism adaptation, highlighting the importance of the cerebellum in minimizing end-point error in reaching.

As predicted by Masao Ito's theory of motor learning [6, 158], the end-point error in reaching is represented by CF signals in the hemisphere of lobule V, the classical anterior lobe arm area of the monkey cerebellum [159]. In this study, liquid crystal shutters eliminated visual feedback during a rapid reaching movement, and two components of CF error signals were identified: a predictive component during the movement, and another originating from visual feedback given at the conclusion of the movement. Notably, CFs form another peak of information regarding the target position after the presentation of the target. In summary, CFs encoded three peaks of information: target position (Peak 1), error in prediction (Peak 2), and end-point error given by vision (Peak 3). These three signals form a temporally ordered cascade: the first conveys goal information suitable for online control, while the latter two provide error information appropriate for driving trial-by-trial adaptation in error-based learning.

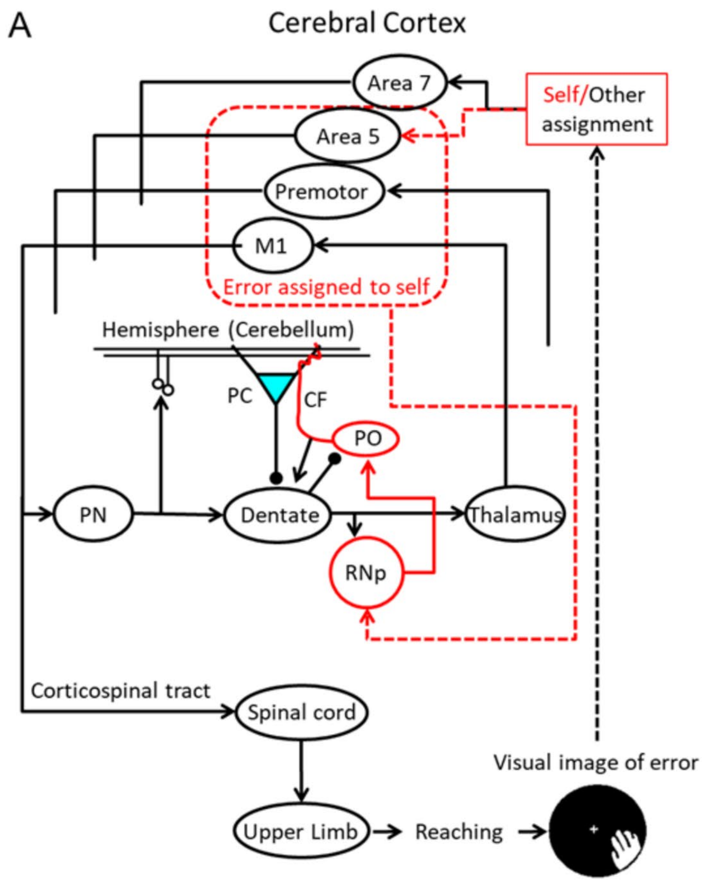
Where does the CF information come from? Since the lateral part of the cerebellar hemisphere receives CF input from the primary olive (PO) and the PO receives its major input from the parvocellular part of the red nucleus (RNp), it is most probable that the CF information originates in the RNp (Fig. 3A). This was confirmed: RNp neurons exhibited three peaks of information similar to CF signals [160] (Fig. 3B). To test the causal link between the error signal and the adjustment of error in reaching, electrical microstimulation was delivered to the RNp just after the end of the reaching movement. Post-movement stimulation induced trial-by-trial increases in error in the direction opposite to the preferred direction of predictive error (Peak 2) and visual error (Peak 3). These findings strongly suggest that CF error signals originating from RNp contribute to minimizing errors in ecological conditions.

We further traced the source of information back to the cerebral cortex, examining M1, the premotor cortex (PM), and Brodmann Area 5, whose stimulation evokes CF responses in the lateral hemisphere of the lobule V [161], as well as Area 7, which does not project to the RNp [162, 163]. We found that information regarding visual feedback-error (Peak 3) was encoded by neurons in all these areas [164, 165]. However, post-movement microstimulation induced increases in error when stimulation was delivered to M1, PM, and Area 5, but not when delivered to Area 7 [164, 165].

We also tested whether Areas 5 and 7 discriminate between errors induced by prism displacement and those induced by a target jump during the movement period [165]. Although the size of visual error — the discrepancy between the hand and the target — was identical in both conditions, Area 5 did not encode the error when it was induced by the target jump. Area 7 alone encoded errors induced by the target jump, demonstrating that the cerebral cortex resolves an error assignment problem: whether the apparent error was due to erroneous motor control (self) or unexpected movement of the target (others). Microstimulation of Area 7 induced gradual adaptation of reaching toward the preferred direction of target jump. This adaptation does not involve the RNp, PO, and the lateral hemisphere of the cerebellum.

Taken together, these results suggest the following likely scenario focused on the Peak 3 visual feedback-error (Fig. 3A): 1) The cause of the visually detected error is assigned to the self (Area 5) or the target (Area 7). 2) The self-assigned visual error in Area 5 is shared with M1 and

A



B

	1: Target (visual)	2: Error (prediction)	3: Error (visual)
Area 7 [15]	++++	±	+++
Area 5 [15]	++++	±	+++
PM [14]	±	±	++
M1 [14]	±	±	++
RNp [10]	+++	++	+++
CF [9]	+++	++	+++

Fig. 3 Sources and temporal organization of climbing-fiber signals in reaching adaptation. (A) The critical pathway conveying visually detected end-point error to the CF is highlighted in red. Note that Area 5, but not Area 7, is involved. **(B)** Summary of three peaks of information represented in key regions. The numbers of plus signs (+) indicate the approximate amount of information. The critical pathway conveying visually detected end-point error (peak 3) to the CF is highlighted in red. Reference numbers are provided in square brackets. Abbreviations: M1, primary motor cortex; PC, Purkinje cell; PO, primary olive; PN, pontine nucleus; RNp, parvocellular part of the red nucleus

PM through rich connections between these areas [166]. 3) The error information is relayed to the RNp, PO, and finally to the lateral hemisphere around lobule V, where plastic changes such as LTD between PFs and PCs achieve adjustments that reduce the error in subsequent reaching movements. The transmission of Peak 3 error signals through M1 is consistent with the feedback-error learning framework, which proposes that feedback-error signals represented in M1 are conveyed via climbing fibers to the cerebellum, where they are used to refine the internal model [167] (see also Chapter 12).

The scenario outlined here generally aligns with Ito's theory of motor learning but leaves several questions unanswered. First, the exact site of adaptation remains unclear: multiple lobules, beyond lobule V, could be involved. This is due to the fact that individual CFs bifurcate into several branches, synapsing with PCs across multiple lobules [168]. Electrical coupling between IO neurons further amplifies the synchronous effect of CFs along a sagittal strip over many lobules [169–171]. In a lesion study on monkeys [154], prism adaptation was abolished only when lesions encompassed not just lobule V, but also Crus I and II, the paramedian lobule and the dorsal paraflocculus of the cerebellar hemispheres and lobule IX. By contrast, partial lesions confined to lobule V and the paramedian lobule did not entirely disrupt adaptation. Second, there is no direct evidence that LTD is involved. Finally, the role of Peak 1 information (target location) remains underexplored. It is possible that Peak 1 information plays a role in online control of reaching through synchronous modification of PC activity, or perhaps it contributes to refining an internal model that predicts where a significant target would appear. CFs may be involved in multiple functions even within the brief window of a reaching movement.

TD Error, Eligibility and Temporal Basis Underlying Cerebellar Learning (Shogo Ohmae and Javier F Medina)

The cerebellum plays a critical role in predicting the time of future sensory events and using that information to precisely control the timing of anticipatory actions [172, 173]. Here, we summarize recent studies that provide new mechanistic insight into how the cerebellar circuit may accomplish this function using temporal difference (TD) learning [174].

TD learning is an algorithm for predicting future events by comparing temporally successive predictions. In the brain, these predictions are learned by a postsynaptic neuron. The equation for updating predictions in real time is given by:

$$\Delta P_t = \alpha \delta_t \sum_{i=1}^{n} e_i\left(t\right) \phi_i\left(t\right) \tag{1}$$

where α is a learning rate constant, and:

$$P_t = \sum_{i=1}^{n} w_i \phi_i\left(t\right) \tag{2}$$

is the prediction at time *t*, computed in the postsynaptic neuron by adding together the activity of each individual presynaptic input at time *t*, $\phi_i(t)$, weighted by its current synaptic strength, w_i,

$$\delta_t = S_t - P_t + \gamma P_{t+1} \tag{3}$$

is the 'TD Error', computed as the difference between the actual, S_t, and predicted, P_t, occurrence of the sensory event at time *t*, and adding to this difference the discounted prediction in the next time step, γP_{t+1},

$$e_i\left(t\right) \tag{4}$$

is the 'Eligibility' of the ith presynaptic neuron at time *t*, which determines how much its synaptic weight can be modified,

$$\phi_i\left(t\right) \tag{5}$$

is the activity of the ith presynaptic neuron at time *t*. For the whole population, the vector $\boldsymbol{\phi}\left(t\right)$ constitutes a 'Temporal Basis'.

TD-Error Signals of CFs

Recent experiments have revealed that the activity of CFs during cerebellar learning tasks displays a hallmark of TD-error signals [28, 146, 147, 175–180]: some CFs respond reliably to sensory events like rewards or eyepuffs when these stimuli are presented unexpectedly ($P_t = P_{t+1} = 0; \delta_t \cong S_t$), but after learning, when the sensory event is being correctly predicted, the same CFs will respond less to it ($S_t \cong \gamma P_{t+1} - P_t$; $\delta_t \cong 0$) and more to any sensorimotor cue that predicts it (at t=time of cue,$S_t = 0; \delta_t = \gamma P_{t+1} - P_t > 0$).

Although these predictive CF responses are consistent with encoding a TD-error signal [123, 174], some questions have been raised. For instance, there is currently no empirical support for one of the key predictions of the TD-error algorithm [123, 181]: that the CF response should occur at the time of the sensory event at the beginning of learning, and then gradually shift earlier over the course of multiple trials until finally emerging at the time of the predictive sensorimotor cue at the end of learning. Others have pointed out that CFs that respond to the cue after learning often display additional responses that are difficult to reconcile with TD-error signals, including: responses to the cue before learning starts [28], movement-related and ramping responses [175, 177], persistent responses to the predicted sensory event after learning [182], and even responses at the time of event omission [146]. More work is needed to fully disentangle CF representations of TD-error from those about 'salience' [180], 'expectation' [147, 178], or 'unsigned prediction error' [146, 147]. Ultimately, the precise timing of the TD-error given by Eq. (3) is likely to differ substantially from task to task, depending on the value of the discount factor, γ, and on the difference between temporally successive predictions [183, 184], which can vary greatly according to Eq. (1) depending on the 'Eligibility' and the 'Temporal Basis' of the presynaptic neurons engaged in each task.

Temporal Basis of GCs TD Learning models assume that the cues that predict upcoming sensory events trigger time-varying activity in the population of presynaptic inputs. The particular profile of the basis functions, $\boldsymbol{\phi}\left(t\right)$, can have a major impact on the precision of the temporal prediction that is ultimately learned [185], as can be intuited from Eq. (2). For instance, maximum temporal specificity can be achieved with 'Complete-Serial-Compound' or 'Delay Line' representations [123, 186], which only allow a single presynaptic input to be active in any given time step after the cue. In contrast, the temporal resolution of the prediction is reduced in 'Microstimuli' or 'Spectral' models in which the activity of each individual presynaptic input peaks at a different time after the cue but partially overlaps with the activity of other inputs [185, 187].

GCs are thought to provide the temporal basis functions necessary for cerebellar learning [172], but there is limited empirical support for this idea partly due to technical difficulties using electrodes to record the cue-evoked activity of large populations of these tiny and densely packed cells [188]. Fortunately, GC population responses can also be examined with optical imaging tools [29, 189–195], and new studies reveal that the activity of many GCs ramps up or down after a sensory cue [29, 182, 190, 196], sometimes for seconds [182, 196]. The onset and slope of the ramping activity is heterogeneous across the GC population, thus providing a complete basis set that can support temporal prediction [182]. Computational studies based on the known local network connectivity of the cerebellar cortex and the intrinsic synaptic properties of its constituent neurons predict that in addition to ramping activity, GCs should be capable of generating basis functions with other temporal profiles [125, 197–201]. But none of these profiles may be discernable in the naïve state because the cue-evoked ramps of GC activity observed in the imaging studies emerged during learning and were absent in the initial stages of training [29, 182, 190, 196]. It is tempting to speculate that the profile of the temporal basis functions may be learned and adaptively tuned according to task demands [182].

Eligibility of Cerebellar LTD

In the standard TD Learning algorithm, eligibility traces often take an exponentially decaying form [174]:

$$\boldsymbol{e}_t = \gamma\lambda\boldsymbol{e}_{t-1} + \boldsymbol{\phi}_t \qquad (6)$$

Equation (6) ensures that when a presynaptic input is activated it becomes eligible for synaptic weight modification, and then gradually loses its eligibility every subsequent time step it is inactive, at a decay rate controlled by the product, $\gamma\lambda$. Thus, when $\gamma\lambda > 0$ the eligibility trace provides a partial solution to the temporal credit assignment problem [174], by allowing updates to the weights of synapses that are active not just at the time of the TD error, but also earlier in time. However, exponentially decaying eligibility traces are suboptimal for temporal prediction in many learning tasks because they cannot guarantee that the synapses contributing most to the TD error are the ones maximally eligible for weight updates.

Recent in vitro studies reveal two ways in which the cerebellum is able to optimize eligibility traces for LTD at the GC-to-PC synapse (GC-PC): In the first solution [202, 203], only those GC-PC synapses that are active at the same time as a 'perturbation' CF signal become eligible for synaptic weight modification, and are allowed to undergo LTD if a secondary 'error' CF signal follows shortly. This plasticity algorithm, known as stochastic gradient descent, assigns credit to the 'right' synapses using the 'perturbation' CF to increase PC activity momentarily, and reducing the strength of co-active GC-PC synapses if the increase in PC activity leads to an error shortly afterwards. In the second solution [204], GC-PC synapses become eligible for modification only at a fixed delay after activation. This delay matches the feedback delay expected for error-related CF signals to reach the cerebellum, which is task-dependent and may be learned through experience [205].

Bottom Line

Current evidence indicates that the cerebellum uses TD learning to predict the time of future events. The underlying error signals, temporal basis and eligibility traces are optimized for each task depending on what the prediction is used for.

From Elegance to Complexity: Understanding Error Processing in the Cerebellar Cortex (Laurentiu S. Popa and Timothy J. Ebner)

The MAI hypothesis is one of the most elegant and enduring theoretical models proposed of cerebellar function [4, 5, 206]. Alas, the cerebellar function seems to abide less by the elegance of simplicity and more by the shimmering demands of complexity.

The doctrine imposed by the MAI model, elegantly coupling the specific cerebellar architecture and physiology to function, states the CS activity exclusively conveys error signals that drives SS adaptation. In this view, CS discharge is necessary and sufficient to drive the SS adaptation, which in turn participates in motor control (behavior expression). Although the MAI model is supported by numerous observations [136, 137, 143, 207–209], a growing body of evidence requires a more nuanced description of the cerebellar function.

CS activity poses a fair number of questions for the MAI doctrine. One major challenge arises from how consistent CSs respond to perturbations and errors. In certain recent experimental designs, the CS modulation is not consistent with an error signal, persisting as behavioral adaptation occurs, while in other experiments behavioral perturbations failed to elicit CS responses altogether [see reviews 210, 211]. Prominent recent examples include multiplexed CS responses during a lever pull but not clear error encoding

[148], lack of CS error modulation during novel visual motor associations [212], and signaling learned associations not errors [146]. These observations suggest that the CS error signals are conditional with the behavioral context as opposed to an unconditional error response. Moreover, SS and behavior can adapt independently relative to CS modulation suggesting that CS response is not necessary for cerebellar learning [212–214]. Also, CS discharge encodes a wide range of behavioral aspects such as movement kinematics [215], occurrence and anticipation of rewards [147, 176, 179] and even salience [180], contradicting the assumption that CS activity is preferentially engaged with error signaling. Importantly, the more studies address the CS error hypothesis, the more exceptions are documented. Furthermore, the MAI hypothesis ignores observations that CF input is essential for maintaining normal cerebellar function [216]. CS activity maintains homeostatic control through history dependent dampening [217] and also maintains homeostatic control of the SS kinematic encoding [218]. The community must not blind itself to only positive findings, if we are to fully understand the role of the CFs in cerebellar function.

Although older studies hinted at a role for SS activity in error signaling [211], recent work established that the SS activity exceeds the classical MAI view that limits the SS function to control either the kinematics or dynamics of the effector movement. In a task requiring continuous processing of motor errors, such as manually random tracking, PC SS activity is robustly modulated not only by kinematics (position, velocity, speed) but also by several measures of performance errors (position, radial, position direction [219]). Moreover, kinematic and performance error parameters are dually encoded as both predictive and feedback representations, consistent with the signals necessary to compute sensory prediction errors [219, 220]. Manipulations of the sensory feedback show that the predictive performance errors signals are based on the predictions of the sensory consequences of motor, while the delayed signals encode the sensory feedback of the performance error measures [221]. Thus, SS activity conforms to a forward model of performance errors independent of the kinematic forward model also present in the SS discharge [222]. Interestingly, the kinematic and performance errors representations are integrated at cell level, suggesting that single PCs operate independent forward models in different domains such as kinematic and performance errors [219].

Performance errors arguably bridge the motor and cognitive domains, as the representation of the behavior goal can be a measure of task understanding and suggests that SS signals can overtly encode aspects of the higher function [223]. Newer studies find that while learning to associate visual cues to left or right hand movement, SS activity encodes the success or failure of the previous trial [212]. These SS signals, operating in the non-motor domain, change as the learning progresses, independently from CS discharge, and are not present during the overtrained behavior. Therefore, these signals are similar to prediction errors that compare expected and actual choice outcome and argue for an explicit prediction error representation in the cognitive domain.

While error processing is a major aspect of the cerebellar function, both in motor and possibly in higher domains, error signaling by PCs involves a complex interaction with behavior as well as between SS and CS. Both SS and CS encode error signals as parts of complex behavioral representations, that integrate errors and non-error parameters, while the role of CS activity in driving SS adaptation is conditional. Therefore, experimental findings have outgrown many of the basic tenets of MAI model.

Cerebellar error processing described here has intriguing similarities to several concepts presented in this Consensus paper. The presence of both kinematic and performance error forward models that can simultaneously predict the sensory consequences and the value of motor commands, that mirrors the cerebral assignment of errors discussed in Chapter 5. This integration of these two forward models can facilitate maintaining the homeostasis of responses consistent with cerebellar reserve described in Chapter 14. Moreover, the presence of prediction errors in the cognitive domain generalizes the predictive function of the cerebellum, consistent with the view presented in Chapter 15. This further opens the possibility that similar forward models operate in non-motor domains, as suggested by Masao Ito in which the prefrontal cortex provides both command and target signals [18, 224], thus detecting, preventing and correcting mismatches between intended and actual outcomes as proposed by the UCT (Chapter 2).

Reward Signals in the Cerebellum: Evidence for Cerebellar Reinforcement Learning? (Marie Hemelt and Court Hull)

Contrasting Models of Cerebellar Learning

Across many cerebellar-dependent behaviors, both neural activity and learned changes in predictive motor output are well-described by the principles of supervised learning [225]. This learning model relies on instructional signals that indicate whether behavior was correctly executed. In the cerebellum, CFs can provide such information, and are thought to establish predictive motor behavior by instructing synaptic plasticity at synapses onto PCs.

In contrast, recent evidence has suggested that neural activity in the cerebellum can resemble features of reinforcement learning during tasks guided by predictions about reward [226]. In reinforcement learning, instructional signals do not directly indicate what actions to take (or not take), but instead guide learning based on trial-and-error exploration to converge on optimal, rewarding actions [186]. Such learning has traditionally been associated with the basal ganglia and instructional signals from midbrain dopamine neurons [123]. However, as discussed in Chapter 1, recent work has shown that these dopamine neurons receive direct input from the cerebellum [64, 65], suggesting that the cerebellum is more tightly integrated into the brain's reward learning system than previously appreciated [27]. Here we will discuss cerebellar reward signaling and the question of whether reinforcement learning models may be appropriate to describe cerebellar function in reward-guided behaviors.

Reward-Related Cerebellar Signaling

Reward signaling has been identified in both major input pathways to the cerebellar cortex, the CF and MF pathways [70] (Fig. 4). Because reward signaling in the brain can serve multiple roles, it is necessary to determine whether these signals conform to the predictions of reinforcement learning [186]. To do so, they must have several key characteristics, including 1) they must act to associate predictive stimuli or actions with rewarding outcomes, 2) They must do so in a scaled, probabilistic manner based on experience and expectation. Current evidence suggests that certain cerebellar reward signals exhibit each of these key requirements.

Evidence from both operant and Pavlovian conditioning tasks has revealed CF signals that share common features with the instructional signals necessary for reinforcement learning, reward prediction errors (rPEs). Specifically, CFs respond to reward in naive animals, and to conditioned stimuli that predict reward in trained animals [147, 176, 179, 227, 228]. Further, these reward responses are decreased or abolished once an unconditioned stimulus is expected [147, 176, 227, 228]. CFs can also encode expected reward size [178], and show scalar responses proportional to presumed reward expectation in some tasks [146]. GCs can also encode reward anticipation, delivery, omission, and better-than-expected reward [29]. However, they may not encode responses to reward-predictive cues in the same manner as CFs [29]. rPE-like signals are also evident in the simple spiking of downstream PCs [212], and both molecular layer interneurons (MLIs) and cerebellar nuclear cells may encode reward anticipation [229, 230].

Overall, CF activity shows the most the commonalities with the instructional rPE signals carried by midbrain dopamine neurons. As discussed in Chapter 6, in some cases CF responses can even directly mirror the properties of TD-errors [28], a learning algorithm shown to be effective for reinforcement learning. However, in other cases, there are notable differences between CF responses and canonical rPE signals. For example, CFs do not appear to encode worse-than-expected reward predictions with decreases in firing. Instead, they appear to encode both better- and worse-than-expected outcomes with increased firing [146, 147], consistent with unsigned prediction errors [231]. Like dopamine neurons, CFs can also respond to novel or unexpected neutral stimuli [28, 176], as well as aversive stimuli and the cues that predict them [28]. However, whether CFs can distinguish between stimuli that predict outcomes of different valence remains to be tested. Moreover, while novelty or other stimulus encoding may complement rPEs in reinforcement learning [232], how these diverse instructional signals may interact to promote cerebellar learning remains unclear.

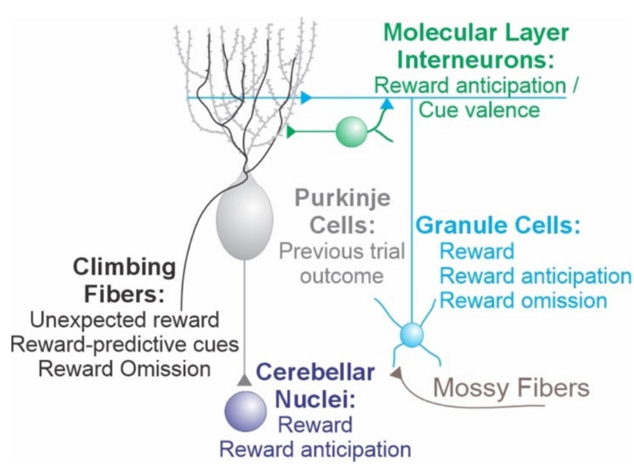


Fig. 4 Reward-related signals across the cerebellum

Cerebellar Participation in Reinforcement Learning

While CF reward signals are similar to reinforcement learning rPEs, the case for cerebellar-dependent behavior that relies on reinforcement learning remains inconclusive. On one hand, data from humans with CA has suggested the cerebellum may learn reward associations from reinforcement, and animal work has indicated a possible role of cerebellar reinforcement learning in learned motor timing and visuomotor associations [73, 227, 233]. In addition, cerebellar projections to midbrain DA neurons have been implicated in pro-social behavior in mice [64]. However, it has remained controversial whether cerebellar damage or disease in humans has a direct impact on reinforcement learning [234–236]. Thus, several questions remain about the role of reinforcement learning as a model for cerebellar processing and learning.

Open Questions

To understand whether and how the cerebellum participates in reinforcement learning, it is necessary to determine how cerebellar reward signals are used for 1) modifying cerebellar output, 2) influencing downstream brain regions, and 3) driving learned changes in behavior. This will necessitate recording from both the cerebellum and its targets during learning, especially those targets known to participate in reinforcement learning. Together with manipulations that can reveal the unique contributions of cerebellar processing, such studies are required to test how these systems work synergistically to support reinforcement learning. Finally, if the cerebellum can support both supervised and reinforcement learning, it will be critical to understand how its largely homogenous crystalline architectures enables such flexibility. Answering these questions will reveal how the cerebellum operates in tasks guided by reward predictions, and how it is integrated into the larger reinforcement learning circuitry of the brain.

Forward Models and the Cerebellum (Reza Shadmehr and Mohammad Amin Fakharian)

The principal ideas that form our current understanding of the cerebellum were generated during the past century in roughly 30-year intervals. Holmes [237] examined soldiers who had suffered gunshot wounds to their cerebellum during World War I and noted that a cardinal feature of their deficit was dysmetria: "The most obvious errors are … toward the end of the movement. As a normal limb approaches its object, its velocity declines at a uniform rate till it comes to rest, but the speed of the affected limb is often unchecked till the object is reached or even passed". Thus, the problem caused by cerebellar damage was not associated with initiating a movement, rather, the problem was stopping the movement on target.

About 30 years later, Marr [4] posited that the IO was the teacher of the cerebellum, and the PCs were its students. Soon after, Ito et al. [124] discovered that PCs were inhibitory neurons with CF regulated learning sites in their PF synapses. Then, 30 years after Marr, Miall and Wolpert [238] posited that the purpose of the CF driven learning was to compute a forward model, i.e., a model that predicted the sensory consequences of motor commands.

Thus, the computational framework that emerged was that cerebellar damage produced dysmetria because the brain could no longer rely on the cerebellum's predictions regarding the relationship between the motor commands and the displacements that they caused. Without the cerebellum's predictions, all that remained was sensory feedback, which, saddled with delays, caused overshooting and endpoint oscillations.

Indeed, during voluntary head movements, PCs as a population appeared to predict the vestibular and proprioceptive sensory feedback [239]. Their firing rates could account for the fact that during active but not passive movements, there was cancelation of the sensory feedback in the vestibular nucleus and DCN [240]. In another example, PCs were modulated during a pursuit task, even when the target was extinguished, suggesting that the cerebellum was predicting the future position of the moving target [241].

However, the problem was that across various types of movements, PC activity continued to be modulated long after the movement had ended [242, 243]. If the cerebellum was critical for control of endpoint accuracy, and the PCs predicted sensory consequences of motor commands, then why were they modulated even after the movement was over?

A key to this puzzle was that the IO monitored the output of the cerebellum and returned to it information that encoded error [159, 217, 244, 245]. This input organized the PCs into anatomical groups called micro-clusters [246–248]. As a result, cerebellar neurons within a micro-cluster learned from a common teacher. The new idea was that the fundamental computational unit in the cerebellum was not an individual cell, but a population of cells that shared a common teacher [249–252].

The input–output structure of the cerebellar cortex resembles a 3-layer neural network, with the MFs as inputs, MLIs as intermediate layers, and PCs as outputs (Fig. 5A). We recorded from the MFs, the MLIs, and the PCs simultaneously as marmosets performed saccades [253] and found that the MFs provided the cerebellum with two pieces of information: a copy of the motor commands $u(t)$, and the location of the target x_g (Fig. 5B). The cerebellar network appeared to integrate the spikes in the motor command MFs to a bound set by the goal MFs. This bound was evident in the constant peak rate that was reached in the population of MLI1s (Fig. 5C). Once the bound was reached, the PCs as a population signaled to the nucleus via disinhibition to produce forces that would aid in stopping the movement (Fig. 5C). However, when the subject made saccades without a visual target (task irrelevant saccades), the target information x_g was missing in the MFs, but a copy of the motor commands $u(t)$ was still present (Fig. 5C, right). Without information about the target location, the PCs could no longer predict when to stop the movement.

Notably, these patterns emerged not in any individual cell, but via an organization of the neurons into groups based on the information in their CF inputs. This input defined the downstream influence of each PC on behavior as a potent vector [254]. By assuming that this potent vector was

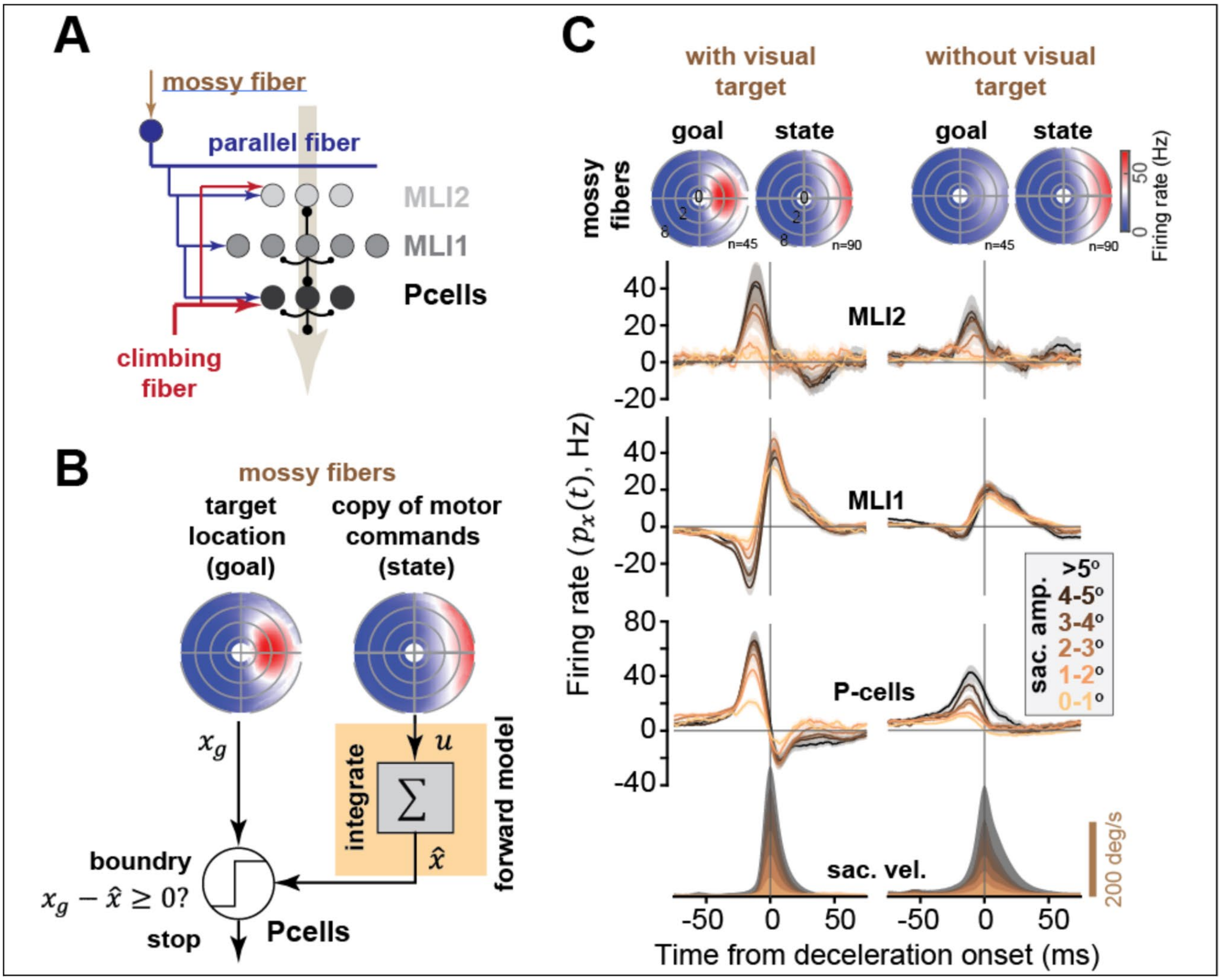


Fig. 5 Output from the oculomotor region of the cerebellar cortex is a command that signals when and how to decelerate an ongoing saccade. **A.** Architecture of the cerebellar network, with MLI2 inhibiting MLI1, and MLI1 inhibiting PCs. **B.** A working model for estimation of the deceleration time based on the goal and motor command information provided by the mossy fiber inputs. **C.** Top: average responses of the goal and state mossy fibers to saccades with and without rewarding visual targets. Bottom, population response of the MLIs and PCs to saccades with and without rewarding visual targets. Note that the MLI1 response reaches the same peak rate regardless of saccade amplitude, potentially reflecting computation of a bound. Adapted from [252]

inherited by all the cells in the neighborhood of the CF, the PC rates exhibited an increase during the saccade acceleration period, then suddenly transitioned to a decrease below baseline at the moment of deceleration onset. The modulations ended as the movement ended.

In order for the PCs to predict deceleration onset, it seems likely that the interneurons in this region of the cerebellum integrated the motor commands $u(t)$ that were conveyed via the MFs to predict displacement $\hat{x}(t)$, which was then compared to the target location x_g provided by the MFs. The computation from motor commands to displacement is a forward model.

Critically, in this region of the cerebellum, the output of the cerebellar cortex, i.e., the PCs, is decidedly not a forward model. Rather, the PCs produce a signal that is more akin to a command that relies on a forward model-like computation that specifies when the movement has reached the target and should be stopped. Damage to the cerebellum produces dysmetria because the PCs are no longer able to predict the exact time when the movement should be decelerated and stopped.

Cerebellar Kalman-Filter Model (Hirokazu Tanaka)

The brain always observes the past state of the body and the world. The world perceived by sensory organs takes some time to process in the brain due to conduction time delay; vision, for example, takes about fifty milliseconds and one hundred milliseconds to reach primary and higher visual areas, respectively [255]. Feedback control based on a delayed state generates oscillatory and unstable movements. The internal model mechanism solves the delayed sensory-feedback problem by neurally implementing the dynamics of a controlled agent [256]. There are two classes of internal models: internal forward and inverse models. An internal inverse model computes feedforward control signal for a given movement trajectory in advance without relying on delayed sensory signals (see also Chapter 12). An internal forward model predicts the current and future states time by time by integrating previously estimated states with an efference copy of motor commands (see also Chapters 9,11, and 15). In physics, equations of motion relate kinematics (i.e., position, velocity, and acceleration) to dynamics (forces and torques). Inverse and forward models solve equations of motion in opposite directions.

Behavioral, neuroimaging, and neurophysiological evidence indicate that the cerebellum, especially the cerebrocerebellum, performs forward-model prediction in motor control, cognitive processing, and social interactions [257, 257, 259, 260]. Deficits in the cerebellum often lead to motor disorders known as CA, reflecting an impairment of predictive computation in the cerebellum [260–262] (see also Chapter 15). The internal forward model plays a role not only in motor control but also in motor learning; motor learning depends not on the performance error (difference between actual movements and target) but on the prediction error (difference between predicted and actual movements) [263]. The cerebellum estimates and predicts for control and learning problems.

The Kalman filter, named after its inventor (Rudolf Kalman), is an optimal estimation that integrates dynamical prediction with noisy observations in the Bayes optimal manner [264]. In the most straightforward formulation (linear dynamical and observation equations with Gaussian noises), the Kalman filter computes a weighted sum of state prediction and observation. A Kalman optimal estimate emphasizes that the dynamics prediction is accurate and observation is noisy, reflecting the statistics of dynamics and observations. The Kalman filter compares the prediction error (technically called *innovation*) between actual observations and predicted observations (observation anticipated from a predicted state). The prediction error, if any, corrects the prediction. Therefore, the cerebellum shares the characteristics of the Kalman filter: predictive computation and learning from prediction errors .

We postulate that the cerebellum is best positioned in the brain to perform the Kalman filter computation (see Fig. 8B in Chapter 15). The cerebrocerebellum receives the projections from the cerebral cortex and peripheral sensory pathways. The cerebellar Kalman-filter model predicts that the cerebellum receives previous state estimates, motor commands from the cerebral cortex, and sensory signals from the sensory pathways. We tested this prediction by analyzing single-unit activities recorded from a monkey performing a wrist-tracking task [265]. The activities included MFs (inputs to the cerebellum), PCs (output from the cerebellar cortex), and DN cells (output from the cerebellum). We found that a weighted sum of the MF activities well reconstructed the PC activities. A weighted sum of the MF and the PC activities well reconstructed the dentate cell (DC) activities. Nonlinear models with quadratic or thresholding terms could not fit the data better than the linear model. The linear equations found empirically corresponded to those of the Kalman filer. The PC and the DN cell equations matched the prediction and the filtering steps, respectively. The crucial prediction of the cerebellar Kalman-filter model was that the current output from the model should predict the future input to the model. We tested and confirmed this prediction by analyzing the population activity data; the DC activities (the cerebellar output) at one time point predicted the MF activities (the cerebellar input) at a future time. We therefore proposed that PCs and DN cells perform the prediction and filtering steps, respectively. Our analysis relied on the data during the motor task. Still, we argue that, based on the known cerebellar anatomy, the same computational principle can be generalized to the cognitive/affective part of the cerebrocebellum [127]. Outputs from non-motor association areas are relayed by the PN neurons and project to the coginitive/affective regions of the cerebrocerebellum. In contrast, inputs to DN originate from a distinct cortical or subcortical sources relayed by non-PN nuclei [266]. Our model predicts that cerebellar cortical lesions impair predictive behaviors and cerebellar nuclear damage deteriorates behavioral adjustments. We conclude that the cerebellar Kalman-filter model provides a unified perspective on the function of the cerebellum in motor and non-motor information processing.

Internal Model of Periodic Sensory Events (Masaki Tanaka, Ken-ichi Okada and Masashi Kameda)

The cerebellum plays a central role in the predictive control of movement and its adaptive learning. During goal-directed movement, the cerebellum receives corollary discharge signals and forms a forward model that internally predicts the motor outcome (see also Chapters 9,10, and 15). Recent findings that the cerebellum is also involved in non-motor cognitive functions have generated great interest in how this traditional view can be generalized (see also Chapter 2). In this section, we present recent physiological studies on the cerebellum, focusing on temporal predictions of periodic events that require internal models.

Predictive Synchronization

The cerebellum is essential for rhythmic movement. Patients with cerebellar damage have difficulty synchronizing their movements to periodic stimuli, especially when the sequence of movements is discontinuous [267]. Synchronized movement requires generation of internal models for periodic stimuli, monitoring of timing errors, and continuous updating of internal models [268, 269]. In general, more areas in the brain are activated during reactive movements to a series of randomly timed stimuli than during synchronized movements to periodic stimuli [270]. However, since some parts of the cerebellum and the frontal and parietal cortices show greater activity during synchronized movements [271, 272], these areas may be involved in the generation of the internal model.

Although predictive synchronized movements to periodic stimuli were thought to be limited to species with vocal learning abilities, such as humans, songbirds, and dolphins [273], recent studies have shown that even monkeys can be trained to generate synchronized movements through reinforcement learning [274, 275]. Neuronal activity in the posterior part of the cerebellar DN, which sends signals through the thalamus to the frontal and supplementary eye fields [39, 276], was examined in monkeys trained to generate saccades in synchrony with periodic alternating left–right visual stimuli [277]. There were three types of neurons with increased activity either before ipsilateral saccades, before bilateral saccades, or immediately after saccades. Of these, neurons active before ipsilateral saccades showed ramping activity that correlated well with movement timing. However, these activities were also observed during reactive saccades to targets presented at random timing, suggesting that these signals regulate saccades independently of stimulus rhythm. On the other hand, neurons active before saccades in both directions showed greater activity during synchronized saccades, with peaks that coincided with the timing of visual stimuli rather than saccades, suggesting that they predict the timing of target appearance (Fig. 6a). Furthermore, post-saccade activity correlated well with the time difference between target appearance and saccades (that is, temporal error) [277]. Thus, neurons in the cerebellum appear to have information necessary for predictive control, including temporal prediction of target appearance, coordination of movement timing, detection of errors, and updating of internal models. Because DN is the output node of the lateral cerebellum, the information is sent to the brainstem and to the cerebral cortex through the thalamus. Future

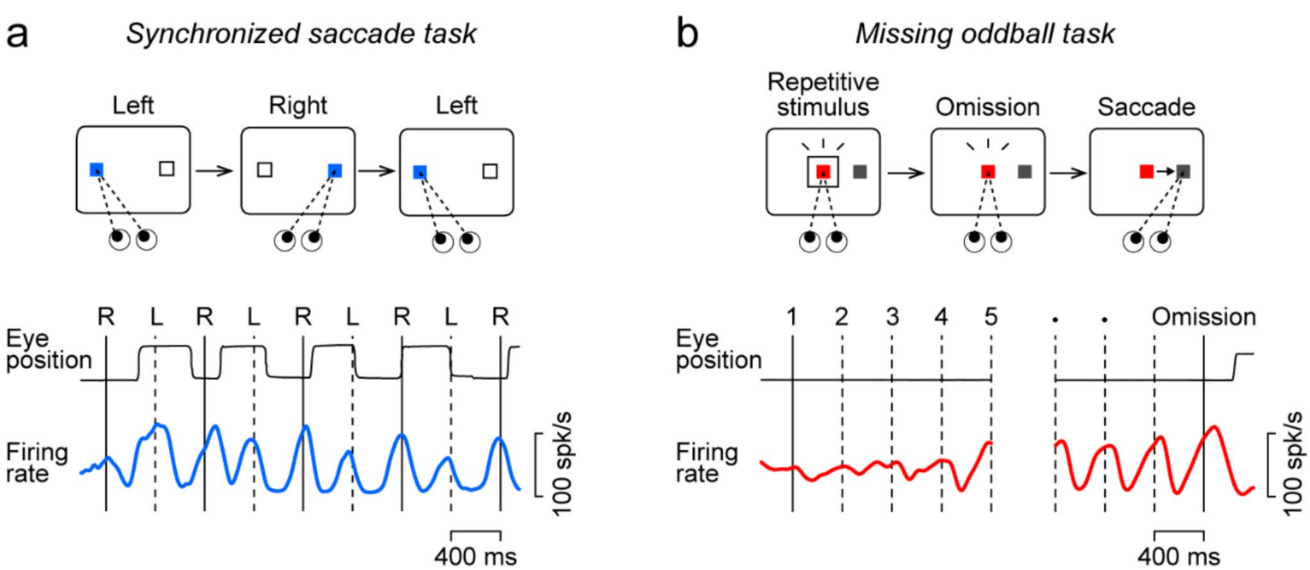


Fig. 6 Single neuron activity in DN of the cerebellum predicts the timing of periodic visual stimuli. a: Synchronized saccade task. b: Missing oddball detection task. In both panels, data are aligned with stimulus onset. Note that neuronal firing rate peaked around the time of stimulus onset in both conditions

studies are needed to clarify how these signals are integrated and converted between multiple cerebello-cortical loops. In addition, it is important to elucidate how the internal model of stimulus timing (or "internal clock" [269]), which is a prerequisite for predictive synchronization, emerges from the interplay between intrinsic cerebellar oscillatory properties [278] and cerebellar learning.

Rhythm Perception

Even in the absence of movement, the brain seems to generate internal models of periodic stimulus timing. When we perceive rhythm in periodic events, such as a musical beat, we can predict the timing of the next stimulus, focus our attention on that moment, and quickly notice slight changes in the rhythm. Rhythm perception involves the cerebellum and basal ganglia in addition to multiple areas in the cerebral cortex [279–281]. Recent studies in monkeys have explored the underlying neural mechanisms [282–285]. In one study, animals have been trained to detect an unexpected omission of periodic stimuli (the "missing oddball" paradigm). In this task, visual stimuli are repeatedly presented around the fixation point at a regular interval, and monkeys respond to the stimulus omission with eye movement (Fig. 6b). To detect stimulus absence, the animals need to learn the stimulus tempo and predict the timing of the next stimulus. Neurons in the cerebellar DN [286, 287], the striatum (caudate nucleus) [288], and the VL thalamus [289] exhibit periodic activity that gradually increases with stimulus repetition. This entrained activity is greatly reduced when the animals attempt to detect a change in stimulus color rather than the stimulus omission, suggesting that these activities are under top-down control and involved in predicting stimulus timing [286, 288–290].

Do these internalized neural rhythms reflect predictions of stimulus timing or periodic preparation of eye movements? To distinguish these possibilities, neuronal activity in the cerebellum and striatum was examined by placing repetitive stimulus and saccade target independently on either side of the fixation point [291]. Many neurons did not change their activity significantly with any combination of stimulus location, suggesting that they have pure temporal information. However, in the population as a whole, DN neurons varied the amplitude of periodic activity depending on the location of repetitive stimulus and caudate nucleus neurons depending on the direction of future saccades. These results suggest that neurons in the cerebellum carry sensory information for repetitive stimuli and those in the striatum carry information related to motor preparation. Thus, the cerebellum appears to generate an internal model of sensory stimuli during rhythm perception, but the mechanism of generation awaits further study.

Cerebellar Internal Inverse Model (Huu Hoang, Hiroaki Gomi and Mitsuo Kawato)

An inverse model is a type of internal model that generates the necessary motor commands to achieve a desired movement goal (see also Chapter 10). In 1987, Kawato and colleagues introduced a hierarchical control model involving the cerebellar forward and inverse models for voluntary movement [292]. This model comprises two main components: feedback and feedforward controllers. The feedback controller, performed mainly by the motor cortex, functions in a reactive manner, modifying motor commands based on sensory feedback. Through learning, the feedforward controller is acquired by the cerebellum guided by error signals from the feedback controller, and produces the necessary motor commands in one-shot. The critical aspect of learning inverse models is the essential transformation of error signals from sensory to motor coordinates. During movement, the error signals represent the difference between the desired and actual trajectories, initially detected in sensory coordinates. The feedback controller converts these errors into motor coordinates, which are then used to train the cerebellum [167]. As the cerebellum refines its computations over time, the motor system becomes more efficient, with the feedforward controller taking on a more dominant role, as well as an improvement of feedback control (Fig. 7A) [293].

To experimentally address whether the cerebellum employs inverse models in a specific motor task, it is essential to examine the nature of the input–output relationship. If the cerebellum's output aligns with movement dynamics in motor coordinates (such as generating motor commands or forces), it indicates the use of an inverse model. For eye movements, Shidara et al., [294] and Gomi et al., [295] have shown that PC input is associated with sensory coordinates, while their output is related to motor coordinates. This pattern indicates that the cerebellum transforms sensory information into motor commands, supporting the inverse model. For upper limb movements, Yamamoto et al. [296] identified PCs in the cerebellum that specifically represent movement dynamics, distinct from those representing movement kinematics. This functional separation supports the presence of inverse models in the cerebellum, as the PCs involved in encoding movement dynamics are closely linked to motor commands. Another line of evidence comes from CF inputs, which were shown to represent error signals in the motor coordinate [159, 164, 297]. In some studies, PC activity has been directly linked to movement execution [229, 298], reinforcing the role of inverse models.

It is worth noting that evidence also suggests the cerebellum employs forward models, which predict the sensory outcomes of motor commands (see Chapters 9 and 11). In their

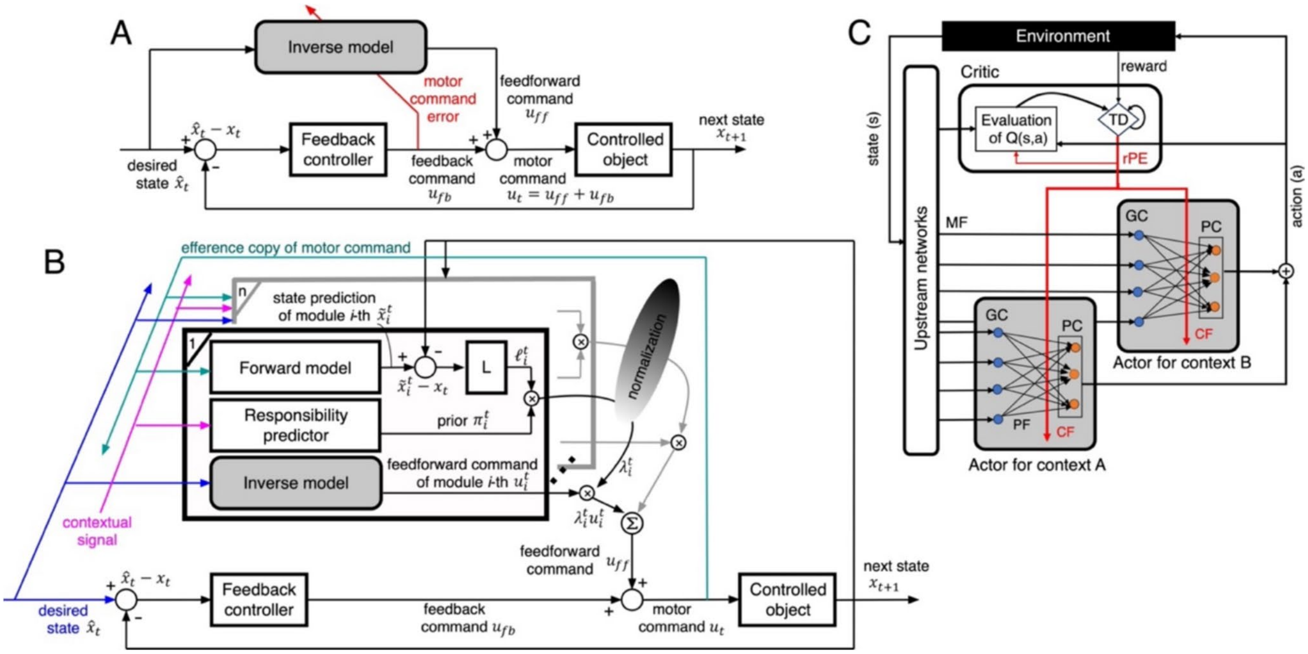


Fig. 7 Cerebellar internal models for sensorimotor learning and reward processing. A: Feedback-error-learning scheme to acquire inverse model capturing the input–output relationship of the controlled object. B: The MOSAIC model with n paired modules, each consisting of a forward model, a responsibility predictor, and an inverse model. The forward model predicts state from motor command, with likelihoods estimated from prediction errors. Contextual cues yield prior probabilities, which are combined with likelihoods to yield the module's responsibility signals λ_i^t. These responsibility signals weight inverse model outputs to form the motor command and scale learning. C: Modular Critic/Actor reinforcement learning in the cerebellum. The critic receives state (s) and action (a) signals, computes a temporal-difference reward prediction error (rPE), and updates both its value function and the actors' policies. It is hypothesized that subsets of Purkinje cells act as context-dependent actors, learning motor commands for different contexts using rPE carried by CF inputs

classical paper, Jordan and Rumelhart proposed that motor learning requires transforming sensory errors into motor command errors, which can be achieved by backpropagating sensory errors through forward models [299]. Accordingly, the cerebellum is thought to acquire multiple pairs of forward–inverse models to coordinate complex motor actions [300]. Haruno and colleagues proposed the "Modular Selection And Identification for Control" (MOSAIC) framework as a computational implementation of such internal models [301]. In this framework, multiple inverse models control the same object under different contexts, while corresponding forward models are trained to choose the most suitable set of inverse models for a specific context (Fig. 7B). Up to date, a substantial body of evidence supports the acquisition of cerebellar internal models across different species, including humans [257, 302], monkeys [303], rats [304], and larval zebrafish [305].

The cerebellar internal models, initially developed for motor tasks, can also be generalized to encompass cognitive functions [224] (see Chapter 2). Interestingly, recent works have suggested a potential role for the cerebellum in reward-based learning tasks [29, 30]. Considering the cerebellum's modular structure [177], it has been suggested that reward processing might also be modular [147]. Recently, Hoang and colleagues analyzed two-photon recordings of CF inputs to 6,000 PCs across 8 Aldolase-C compartments in Crus II during Go/No-go auditory discrimination tasks [306]. They employed a computational reinforcement learning framework, specifically Q-learning, to accurately reproduce the mice's licking behavior, thereby estimating key reward-related variables—such as reward prediction and reward prediction errors (rPEs)—on a trial-by-trial basis. Subsequently, through advanced regression analyses, they revealed representations of both reward-related and sensorimotor variables distributed across multiple cerebellar modules [150]. Notably, CF activity was found to be negatively correlated with *signed* rPEs, decreasing after positive "better-than-expected" outcomes and increasing after negative "worse-than-expected" outcomes. During learning, both positive and negative rPEs diminished in magnitude, while CF activity shifted in the opposite direction—increasing with positive rPEs and decreasing with negative rPEs. By contrast, other studies suggest that CF inputs may encode *unsigned* rPEs, with activity enhanced regardless of outcome valence (see Chapter 8). Such discrepancies likely arise from the cerebellum's modular organization, where distinct modules may implement different coding schemes [147, 177], as well as from differences in task design (e.g., discrimination tasks such as Go/No-go versus conditioning paradigms). Interestingly, the spatial distribution of these modules showed a remarkable alignment with the Aldolase-C expression patterns in the cerebellar cortex [307]. Despite this apparent alignment at the broader scale, the authors also observed that individual modules were often intermixed within a single cerebellar zone—and in some cases, even within single Purkinje cells—suggesting a finer organization that transcends classical Aldolase-C-defined boundaries [306]. The coexistence of overlapping modules within the same anatomical domain points toward a flexible computational framework, in which discrete functional modules can independently learn and adapt to task-specific demands while maintaining coordinated behavior through shared anatomical structure [308]. While still speculative, these findings suggest that the cerebellum may develop an internal model specifically suited to support modular reinforcement learning. This model likely involves a network loop that includes the IO nucleus, cerebral cortex, basal ganglia, and cerebellum. In this framework, the cerebellum receives predictive and negative rPE signals to guide motor behaviors, acting as a specific-context actor in reinforcement learning (Fig. 7C). Supporting this hypothesis, a spiking neural network of the cerebellum—with a modular architecture and bidirectional plasticity at the parallel fiber-Purkinje cell synapses—has successfully reproduced both behavioral outcomes and neural activity observed in Go/No-go task [150]. However, the mechanisms underlying interactions between the cerebellum and other brain areas in such modular reinforcement learning remain to be explored.

Artificial Recurrent Neural Circuit of the Cerebellum for Replicating Cerebellar Language Functions (Shogo Ohmae And Keiko Ohmae)

The cerebellum plays an essential role in various cognitive functions, including language (see also Chapter 2), in addition to its well-known involvement in motor control (see also Chapters 1,3,5,7,9,12, and 15). However, the circuit computations underlying its cognitive language functions remain largely unexplored. The cerebellar language functions include both motor and non-motor cognitive aspects [309, 310]. This article focuses on the non-motor cognitive language processing, but first introduces both aspects. The motor language function of the cerebellum is related to speech and vocalization, controlling the oropharyngeal vocal apparatus via the articulatory muscles. Neuroimaging studies show that this function involves the bilateral medial lobule VI, corresponding to the facial, tongue, and lip areas of the cerebellar sensorimotor body map. Damage to these areas causes dysarthria. In the DCN, the rostral and dorsal regions of the bilateral DN control vocalization and articulation [311]. In contrast, non-motor cognitive language processing involves the right lateral cerebellum (lobule VI, Crus I/II) and the ventral and caudal right DN. These regions connect closely to the language centers of the left neocortex [312], and the right cerebellar lateralization is thought to depend on the left neocortical lateralization [313]. These regions handle non-motor cognitive language functions such as language fluency, prediction, verb generation, grammar processing, semantic judgment, and linguistic working memory.

The circuit computations in the cerebellum underlying non-motor cognitive language processing remain unclear. As language processing is characteristic particularly in human cognition and cannot be studied through animal experiments, neuronal-level recordings that could shed light on these circuit computations are unavailable for the cerebellum. To address this, we created an artificial neural circuit modeling the cerebellar circuit structure, including the connection patterns of different cerebellar cell types. Additionally, we implemented biologically plausible inputs, outputs, and learning mechanisms to replicate the cerebellar language functions [314]. [Circuit Structure] While the cerebellum is often seen as a feedforward circuit, recent studies have shown that there are abundant feedback pathways from the output region, the DCN, to the input GCs. These include both direct projections [315–318] and indirect projections through the PN or brainstem [319–322], which are essential for cerebellar predictive function [318, 323, 324]. Based on this, we created a three-layer Recurrent Neural Network (RNN) that modeled the cerebellum, connecting input cells, PCs, and output cells through both feedforward and feedback pathways. [Input–Output of the Circuit] The right lateral cerebellum (Crus I/II) is involved in predicting the next word in a sentence (Fig. 8a) [325–328]. During this process, the cerebellum is considered to sequentially receive the word input and output the prediction of the next word [325, 326, 329], and we adopted this proposal to configure the input–output for our artificial cerebellar circuit. This setup aligns with the internal model theory, which states that the central function of the cerebellum is "to create internal models of the world in order to predict future events," and with its extension to language processing. [Learning Mechanisms] We followed the widely accepted proposal that the IO, which provides essential information for cerebellar learning, calculates prediction errors, and assumption that the IO compares the predicted next word (cerebellar output signal) with the actual next word to calculate the prediction error, which is fed back to the cerebellum to update synaptic weights and improve future predictions [129, 329].

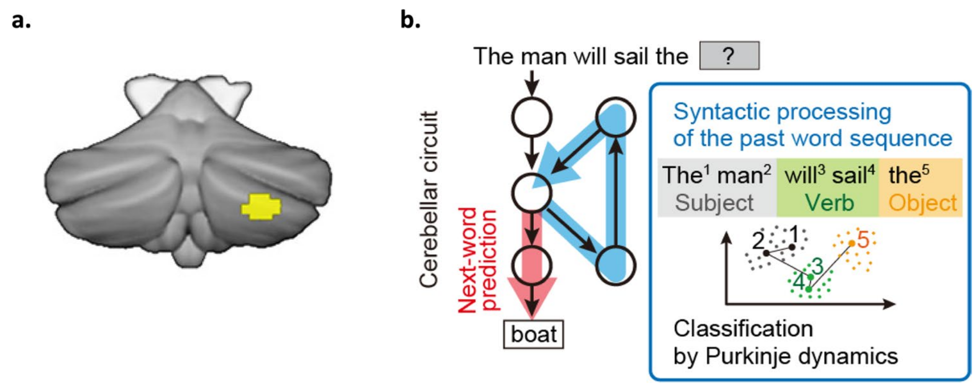


Fig. 8 Artificial neural network imitating the cerebellar circuit structure and language functions. a, Activation of right lateral cerebellum (Crus 1/2) during the next-word prediction, measured by fMRI. Not only the correlation, but also the causal relationship was demonstrated by brain suppression using transcranial magnetic stimulation (TMS). Figure adapted from [325] (CC BY 4.0).b, Single circuit computation achieved both next-word prediction and syntactic processing. When the artificial neural network imitating the cerebellar circuit was trained by next-word prediction, the output was trained to generate predictions for the next words (red arrow). Interestingly, upstream of the predictive output neurons, the syntactic processing circuit (blue arrow) spontaneously emerged so that the PCs can represent syntactic information (by classifying subject, verb, and object words). Figure adapted from [314](CC BY 4.0)

As a result, when the artificial cerebellar circuit was trained for next-word prediction, it not only acquired the ability to predict the next word (Fig. 8b, red arrow), but also the intermediate layer of the word prediction circuit (PCs) spontaneously developed the syntactic processing ability, another language function of the cerebellum [309, 330–333] (Fig. 8b, blue). As previously mentioned, next-word prediction aligns with the internal model theory, a core function of the cerebellum. Among the language tasks involving the right lateral cerebellum (such as language fluency, prediction, verb generation, grammar construction, semantic judgment, and working memory), the next-word prediction function likely contributes to language fluency, prediction, and working memory (note that the feedback pathway stores past word information for prediction). On the other hand, syntactic processing aligns with the sequence processing theory, which proposes another key function of the cerebellum. Sequence processing is essential for functions such as grammar construction and semantic judgment (see the list above). This suggests that the next word prediction and syntactic processing are the core of language functions of the cerebellum. Traditionally, these two functions have been viewed as fundamentally distinct. However, our cerebellar model, represented by a three-layer RNN circuit that predicts the future word based on past word inputs and learns from the prediction error, has demonstrated that both functions can be explained in a unified manner as the two outputs of the single circuit computation. While efforts to understand human cognitive functions through the biologically plausible artificial neural circuits is still in their early stages, this promising field is expected to grow, leading to exciting discoveries in the future.

Cerebellar Reserve and the Cortico-DCN Loop Model (Hiroshi Mitoma and Mario Manto)

Definition and Characteristics of Cerebellar Reserve

Cerebellar reserve is defined as “the capacity of the cerebellum to compensate and restore function in response to pathologies” [334]. The concept was originally proposed to account for differences in the susceptibility of the cerebral cortex and basal ganglia to the aging process or pathological damage [335–337]. Basically, the reserve moderates pathology and influences outcome [335]. On the other hand, the concept of cerebellar reserve places more emphasis on the capacity of the cerebellum to restore lost function and resist to pathological damage [334]. These salient features date back to the classic paper of Sir G. Holmes [338]. In the case of acute and localized lesions, the surrounding neural circuits are responsible for compensation and repair (referred to as structural cerebellar reserve), whereas in the case of chronic and diffuse lesions, the compensation and repair processes occur in the affected area (referred to as functional cerebellar reserve) [334].

Neural Mechanisms of Cerebellar Reserve: “the Cortico-DCN Loop Model”

One of the neural mechanisms underlying cerebellar reserve is redundancy and diverse synaptic plasticity in the cerebellar cortex [339]. This is explained by the unique anatomical properties of the cerebellar circuitry in terms of cellular connectivity and geometrical structure. Approximately 60–80% of the whole brain's 85–100 billion neurons are located in the cerebellum, although the cerebellum represents only about 10% of the brain mass [340, 341]. In addition, the branching patterns of individual MFs, which carry signals from the cerebral cortex and periphery, are highly divergent, especially along the mediolateral axis [342] and, therefore, MF inputs are highly convergent to each microzone, the functional unit in the cerebellar cortex arranged in a rostro-caudal direction [339]. The PF axons of the cerebellar granular cells form approximately 180,000 synapses with the dendrites of a single PC, though the majority of these synapses (approximately 85%) are silent [343]. It is assumed that the multiple forms of synaptic plasticity based on such redundancy of neuronal numbers and inputs can functionally reconstruct lost functions [339]. It is noteworthy that cerebellar reserve requires the proper functioning of the cerebellar cortex and the DCN. Clinical studies have shown that lesions affecting the DCN are not fully compensated for at any developmental age [344, 345].

The cooperative function of the cerebellar cortex and the DCN can be seen in the neural mechanism of the internal forward model, i.e., state predictor, based on Kalman filter computation. The Kalman filter model characteristically divides the predictive control of movement into two functions: a prediction process and a filtering process, which integrates predictions and peripheral feedback [265]. Tanaka et al. (2019)[265] have demonstrated that the cerebellar cortex is responsible for the prediction process of the Kalman filter model, while the DCNs are responsible for the filtering process, based on the neural activity of PCs, MFs, and DCN neurons recorded during hand step-tracking movements in monkeys [265] (see also Chapters 10, and 15).

Evidence suggests a bidirectional interaction between online predictive control and learning. Continuous predictive control, crucial for accurate movement, appears to facilitate learning by updating the internal forward model through repeated cycles of prediction and error correction, as demonstrated by Tseng et al. (2007) [346]. Therefore, we propose the "cortico-DCN loop model" as a model of cooperation between cerebellar cortex and DCN. This model assumes that the activity of the neural network determined by the Kalman filter model also operates in the cerebellar reserve [347] (see also Chapters 10 and 15). Accordingly, the cerebellar cortex performs predictive computations, and the DCN filters the predictions and peripheral feedbacks to reconstruct lost functions [347].

Therapeutic Strategies based on Cerebellar Reserve

Figure 9 illustrates the relationship between disease progression and cerebellar reserve, as well as the severity of CAs (CAs; Fig. 9). The ordinate represents the level of residual cerebellar reserve, which is closely tied to the severity of CAs. This correlation is underpinned by the shared neural mechanisms governing cerebellar reserve and online prediction control, both of which operate under Kalman filter theory (see also Chapters 10 and 15). Clinically, the severity of CAs is categorized into four stages based on the extent of clinical manifestations: asymptomatic stage, prodromal stage, symptomatic restorable stage, and symptomatic non-restorable stage.

Figure 9 shows two pathological conditions [334]: (1) cerebellar reserve decreases with disease progression, and (2) there is a threshold for cerebellar reserve. Crossing that threshold coupled with loss of cerebellar reserve leads to failure of recovery of cerebellar reserve and CAs, which will remain unchanged even after halting the disease progression process. The above findings were described in a state-of-the-art study on the post-treatment outcomes of various immune-mediated CAs (IMCAs) [334]. Thus, as with IMCAs, even if we use therapeutic agents that are known to directly control the degenerative pathological changes in the cerebellum, the effects of etiology- and pathogenesis-based therapies will be limited by the constraints imposed by the cerebellar reserve. Two strategies can be employed to address this problem. First, treatment should be initiated while there is still a remaining cerebellar reserve [348]. We have already emphasized the importance of early diagnosis and treatment by promoting the phrase "time is cerebellum" [349]. Atrophy on MRI or CT scans can be a marker of cerebellar reserve. However, there are many cases in which reserve is lost even in the presence of only mild atrophy. Therefore, recent years have witnessed extensive search for pathology-related biomarkers that can detect early stages of the disease process (asymptomatic and prodromal stages) to help guide treatment decisions. Candidate molecules, such as neurofilament light chain (Nfl), have been identified recently [350]. On the other hand, from the perspective of cerebellar reserve, it is preferable to directly detect abnormalities in the operation of neural circuits, since the cerebellar reserve is formed by neural circuits.

Second, treatment modalities that can strengthen the cerebellar reserve itself should be introduced [348]. These include rehabilitation, non-invasive cerebellar stimulation (NICS), and neurotransplantation. In fact, the beneficial role of motor rehabilitation in enhancing cerebellar plasticity to alleviate CAs was reported more than a decade ago [351]. Similarly, one of the mechanisms by which NICS ameliorates CAs is thought to be the induction of cerebellar cortical plasticity and the long-lasting changes in the cerebellar control of the cerebral cortex and spinal cord [352, 353]. Neurotransplantation, a promising future treatment, remains currently very challenging to reconstruct and replace complex cerebellar neural circuits [354].

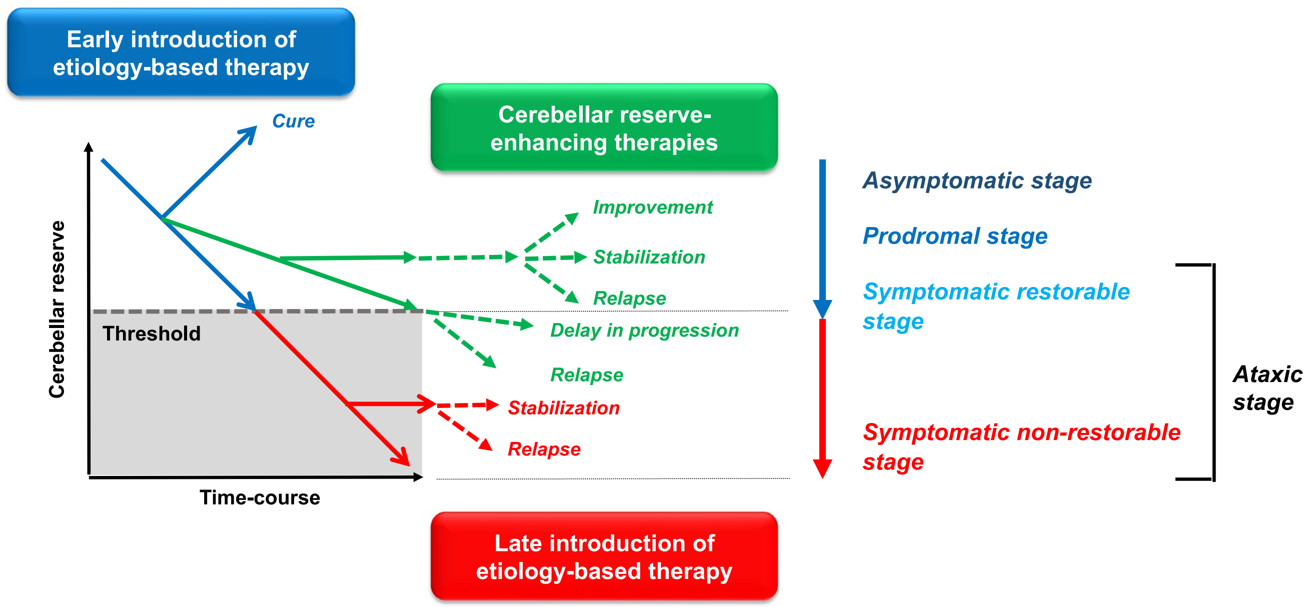


Fig. 9 The concept and relationship between the restorable stage and cerebellar reserve. The clinical course is divided into four stages: asymptomatic stage, prodromal stage, and ataxic stage (subdivided in symptomatic restorable stage and symptomatic non-restorable stage). Even in cases where etiology-based therapy is not feasible, if the patient presents in a restorable stage, cerebellar reserve-enhancing therapy may be considered to improve, stabilize, or delay the progression of cerebellar ataxias (CAs). Extra-cerebellar structures contributing to the compensatory are not shown

Rather, neurotransplantation is expected to prevent neuronal damage and promote plasticity [354]. In support of such scenario, one study reported that transplantation of mesenchymal stem cells into the cerebellum of newborn Lurcher mice produced neurotrophic factors, such as brain-derived neurotrophic factor (BDNF), which can potentiate glutamatergic and GABAergic transmission [355]. These therapies can counteract the decline in cerebellar reserve as the disease progresses and increase cerebellar reserve [348]. The gentle slope or cessation of the progression curve in the Figure represents the effects of therapies on the cerebellar reserve.

The Cerebellum as a General Prediction Machinery (Shinji Kakei and Takahiro Ishikawa)

In our recent studies, we provided neural evidence that the motor part of the cerebrocerebellum predicts future motor cortical activities from present MF inputs in a manner compatible with a Kalman filter (a special type of internal forward model)(Fig. 10A)[127, 265, 356, 357](see also chapter 10, and Figs. 1 and 2 in a review by D'Angelo et al. [358]).

A Kalman filter consists of two steps and features a unique corticonuclear microcomplex (CNMC) shown in Fig. 10B (for various CNMCs, see a review by Apps and Garwicz [315]). The first prediction step performs the predictive computation using efference copy inputs (*IN1*, *MFa*). This step is located in the cerebellar cortex (*CBXa*), and the activities of the PCs encode its output (*Prediction*) [265]. The second filtering step is located in DN. This step integrates *Prediction* with the latest feedback inputs (*Filtering*)(*IN2*, *MFb*)[265]. Regarding the prediction by the Kalman filter, the prediction step in *CBXa* plays the primary role, and the filtering step in DN plays the secondary role.

We also found that the timing of the DN activities in the cerebrocerebellum was consistent with the cerebellar Kalman filter hypothesis [266, 357]. The DN activities *precede* the actual movement by about 90 ms. The lead times of the DN modulations were slightly earlier than those of the task-related muscle activities [357]. On the other hand, the DN activities *follow* the activities of M1 and the ventral PM [266, 357]. Overall, the DN activities appear to *predict* the future state of the motor apparatus rather than the motor command or the feedback signal. Namely, the DN activities in the cerebrocerebellum are consistent with the cerebellar Kalman filter (or, more generally, forward model) hypothesis regarding timing, representation, and transformation of activities.

Following the prediction of the cerebellar forward model hypothesis, we confirmed impaired predictive control in patients with CA [359–361]. First, CA patients manifested *increased errors in velocity control* for a visually-guided pursuit movement (Fig. 5 in [361]). Second, CA patients also exhibited an increased delay in the predictive component

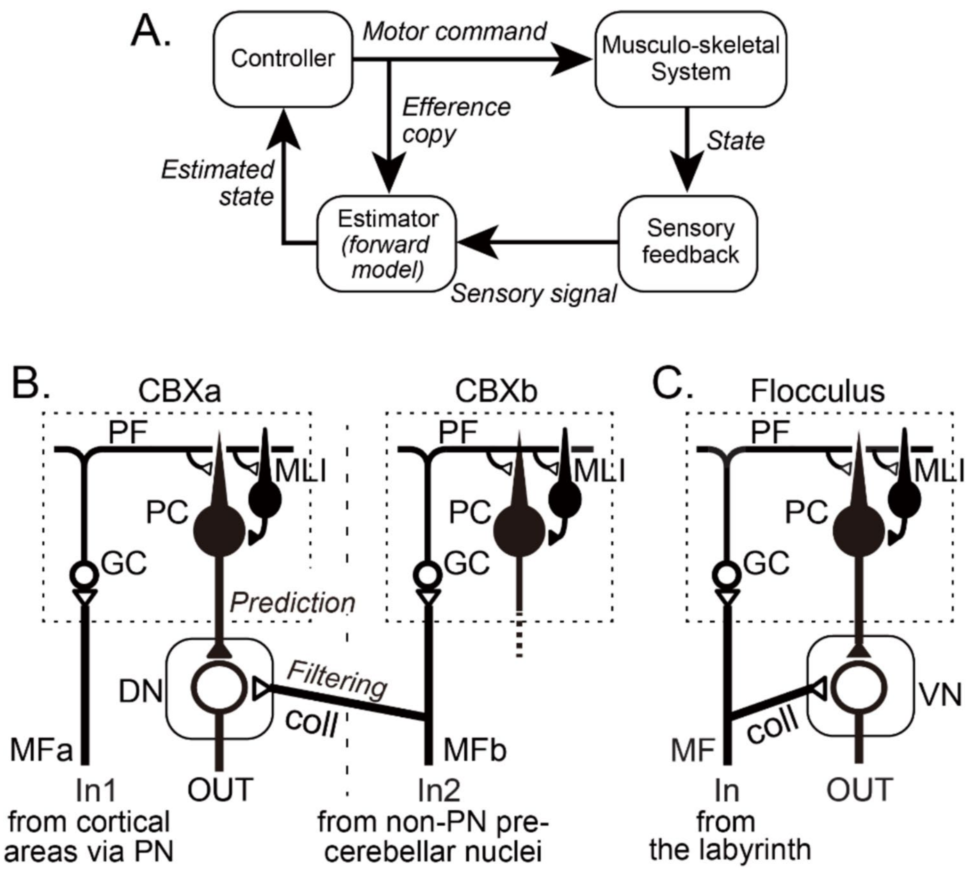


Fig. 10 The forward model and its implementation in the cerebrocerebellum. (A): Schema of closed-loop optimization. The Estimator (forward model) receives the Efference copy of the Motor command and the delayed Sensory feedback (Sensory signal) to compensate for the feedback delay. Note that the Estimator and the Controller form a closed loop. Thus, they continue to generate motor commands even when sensory feedback is unavailable or unreliable. (B) and (C): Two types of cortico-nuclear organization. (B) The cortico-nuclear organization compatible with the forward model in (A). In this scheme, MFs (MFa) from pontine nuclei (PN) project to the cerebro-cerebellum (CBXa) without collaterals to DN, whereas other MFs (MFb) project to DN with collaterals. Note that MFa and MFb have distinct origins and projection areas in CBX. (B) is consistent with the latest morphologic data for the cerebrocerebellum. (C) The textbook scheme of cortico-nuclear organization for the flocculus. It is incompatible with the forward model in (A). In this scheme, the same MF from the labyrinth projects to the flocculus and VN with collaterals (coll). CBX, cerebellar cortex; coll, collateral; DN, dentate nucleus; GC, granule cell; In, input; MF, mossy fiber; MLI, molecular layer interneuron; PC, Purkinje cell; PF, parallel fiber; PN, pontine nuclei. Modified from [127] under CC-BY license

of the pursuit movement. In the control subjects, the predictive component lagged the target motion by only 66 ms, which was too short for a visual feedback delay [361]. The slight delay provided proof of the predictive nature of the component. In CA patients, however, the delay increased by more than 100 ms, as much as 172 ms. The increased delay was comparable to a visual feedback delay, proving a lack of compensation for sensory feedback delay due to the impaired forward model. In summary, we confirmed the impairments of the forward model regarding accuracy and delay of state prediction in CA patients.

The predictive nature is not the monopoly of the cerebrocerebellum, the newer part of the cerebellum. In the flocculus, an older part of the cerebellum, the neuron circuitry of the CNMC (Fig. 10C) is *inconsistent* with the forward model (or Kalman filter), where PCs and vestibular nuclear cells (VNCs) share the common MF input. Nevertheless, the flocculus PC input to VNCs is essential for generating *a phase advance* in VNC activities (p198 in [138]). Note that the phase advance is critical to maintaining a stable gaze by minimizing the effect of the head motion in a *feedforward* (i.e., predictive) manner (Fig. 9A in [138]).

The flocculus also plays a crucial role in the feedback control of smooth pursuit eye movements by minimizing pursuit errors. We use the term "feedback" because its *output* to the extraocular muscles *directly* influences the retinal position of the target (i.e., the *input*). Note that the performance of the smooth pursuit depends on the *prediction* accuracy of the target motion and the eye movement. This way, the prediction prevails even when the cerebellum contributes directly to motor control (see also Chapter 12).

Considering the well-known "crystal-like" regularity of the cerebellar cortical neuron circuitry and the predictive computations in the cerebellar cortex already discussed in Chapters 1 through 15, we propose that the entire cerebellar cortex operates as *prediction machinery*. The outputs from the cerebellum facilitate various *predictive* operations in extra-cerebellar targets, regardless of the origins or modalities of MF inputs and variations of connectivity in the CNMC [138]. The cerebellum serves almost the entire brain, including the cortical areas, the brain stem, the basal ganglia [26, 27], the limbic system, the hypothalamus, and the autonomic nervous system [31–33]. The cerebellar prediction machinery generates automatized parallel outputs unconsciously to maximize the organism's survival by optimizing its fitness to the environment. For instance, it helps to maintain homeostasis by coordinating the autonomic nervous system. It helps to generate dexterous actions by organizing the segmental movements. It helps to organize a complex behavior by arranging an optimal order of actions. It helps to optimize behavioral outcomes by employing more rewarding tactics based on experience.

Discussion

In this Consensus paper, the panel of experts tried to provide updates for novel system-level models of cerebellar function based on advances of the last decades. In other words, we thought it was premature to propose the grand unified theory of the cerebellum based on sound experimental evidence. Therefore, we focused on the cerebellar models that interpret the cerebellar function in the brain as a system and did not suggest a general unified theory for the entire cerebellum. On the other hand, there is a rich literature attesting formidable attempts to resolve the structure–function-dynamics relationship of the cerebellar circuit [e.g., 34, 199, 201, 278, 315, 358]. This issue pertains to more mechanistic models of cerebellar local circuits rather than system-level models. In this paper, we chose to lean towards the more system-level models. Although both types of models are equally essential to get to our ultimate goal, the grand unified theory of the cerebellum, there remains a gap between the mechanistic-level models and the system-level models. To connect the two levels, we need to identify neural encoding for each element and its transition through the cerebellar neuron circuitry in behaving animals. For the time being, we can present some hints towards *the unified theory for the cerebellar cortex,* encouraged by the crystal-like uniformity of the cerebellar cortical neuron circuitry (for instance, the universal cerebellar transform (UCT) in Chapter 2, or "the prediction machinery" in Chapter 15). It should also be noted that an artificial cerebellar circuit model (Chapter 13) may provide a promising way to analyze both representation and information transformation in the cerebellum especially for higher brain functions.

In the following, we are going to focus on the seven points that are related to system-level models of the cerebellum.

Expansion of cerebellar functional domains
Morphological consistency of the cerebellar cortex and functional variety of CNMC
Analyzing the CS activity and its origin
Divergence of the CS activity
The duality of CS-induced plasticity in the cerebellar cortex
CS-induced effects on DCN output
Application of cerebellar models to elucidate CA

Expansion of the Cerebellar Functional Domains

For many years, in the era of the classical MAI model, the cerebellum was believed to regulate the specific aspects of motor coordination. However, recent advances in neuroanatomical (Chapter 1), neurophysiological (Chapters 7 and 8), and fMRI (Chapter 2) studies have added a number of brain regions that have close relationships with the cerebellum. The target of the cerebellar output includes the brain stem, the hypothalamus, the limbic system, the basal ganglia (which includes the dopamine system)(Chapters 1,and 8), the association cortices, as well as the sensorimotor cortices (Chapters 2,5,6,7,9,10,11,12, and 15). In particular, the basal ganglia, the newest member of the cerebellar club, were thought to be independent of the cerebellar function. Nevertheless, recent evidence from both operant and Pavlovian conditioning tasks has revealed climbing fiber signals that share common features with the instructional signals necessary for reinforcement learning, and rPEs (Chapter 8). For instance, climbing fibers respond to reward in naive animals, and to conditioned stimuli that predict reward in trained animals [147, 176, 179, 227, 228]. It is evident that the cerebellum is involved in all levels of neural function, ranging from the lowest level of autonomic functions to the highest level of cognitive-affective functions.

Morphological Uniformity of the Cerebellar Cortex and Morphological Diversity of CNMC

The diversity of the cerebellar functional domains, including motor control (Chapters 7,9,10, and 12), autonomic functions [31–33, 138], and cognitive-affective functions (Chapters 2,8,11, and 13), appears to require a specialized input–output transformation for each domain. On the other hand, if we focus on the cerebellar cortex, its structure is so uniform that it is often described as "crystalline," and a standard conversion rule is assumed in the cerebellar cortex regardless of the region (Chapters 2,7, and 15). Shcmahmann's UCT is an example (Chapter 2). On the other hand, there are site-specific differences in the input–output structures of CNMCs [138](Chapter 15). A CNMC is defined as a set of “a cortical microzone and an associated small group of DCN cells dedicated to a single function” (p. 198 in [138]). While the microzone of the cerebellar cortex is characterized by crystalline uniformity, the interface circuits of CNMCs, i.e., the pattern of MF and CF inputs, show marked local differences. For instance, Ito pointed out six examples of CNMCs based on the difference of main MF inputs to the cortical microzone and of collateral MF inputs to the DCN cells (Fig. 9 in [138]). Overall, a CNMC has a uniform microzone but achieves a specific function due to the variation of its peripheral circuits.

Decoding CS Activity and Identification of its Source

In the MAI model, CS activities were assumed to convey instruction (i.e., error) signals for cerebellar learning. Thus, CS activity in behaving animals has been recorded under various experimental set-ups, such as goal-directed movements [357, 362–364], ocular-following movement [297], saccade eye movement [365], stepping movement [366, 367], or eye-blink conditioning [28]. In these studies, CS activity demonstrated task-specific patterns of modulation. For example, CS activity that coincides with the onset of reaching movement was repeatedly observed in NHP [357, 362–364]. This type of CS activity was observed in successful trials where animals were overtrained with minimal reaching errors. So, it was not amenable to a simple interpretation of error or failure. Kitazawa et al. [159] advanced the analysis of CS activity to the next stage. They reported that single CFs encode three distinct components of information in different time periods during reaching movements: Peak 1: target position; Peak 2: error in prediction; Peak 3: endpoint error given by vision (Chapter 5). In retrospect, the CS activity at the movement onset in the over-trained NHPs [362, 363] corresponds to the Peak 1 component (target position). In the same year, Kobayashi et al. [297] (Chapter 12) also reported that CSs conveyed two components of information (sensory and motor) during ocular following movement. These results suggest that the IO (i.e., the origin of CFs) neurons receive convergent inputs from multiple sources (Chapter 5).

Next, it is vital to prove the causal relationship between the CS signals and the adjustment of movement errors (Chapter 5). Inoue and Kitazawa [160] (Chapter 5) traced the three error signals to RNp and further back to motor areas (areas 4 and 6) and parietal areas (areas 5 and 7). Furthermore, they confirmed the causal link from the CS signals to the adjustment of error in reaching by stimulating RNp just after the end of the movement to provide *artificial* error signals. A similar multiplexed CS signal was also reported during the saccade eye movement in NHP [365]. In this experiment, PCs receive a multiplexed climbing fiber input that merges complementary streams of information on the behavior, separable by the recipient PC because they are staggered in time.

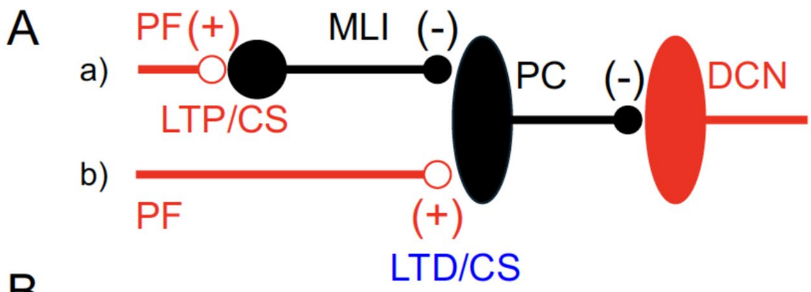
Divergence of CS Activity

CS discharge encodes a wide range of behavioral aspects. Importantly, the more studies address the CS error hypothesis, the more exceptions are documented (Chapter 7). For instance, the cerebellum plays a critical role in predicting the time of future sensory events and using that information to precisely control the timing of anticipatory actions (Chapters 6, and 7) [172, 173]. Recent experiments have revealed that the activity of CFs during cerebellar learning tasks displays a hallmark of TD-error signals (Chapters 6 and 7)[28, 146, 147, 175–180]. Furthermore, recent evidence has also suggested that neural activity in the cerebellum can resemble the predictions of reinforcement learning during the tasks guided by reward predictions (Chapters 7 and 8)[226]. Such learning has traditionally been associated with the basal ganglia and signals from midbrain dopamine neurons [123]. Nevertheless, recent studies have demonstrated that the dopamine neurons receive direct input from the cerebellum [63, 64], suggesting that the cerebellum is more tightly integrated into the brain’s reward learning system than previously assumed [27]. In the field of machine learning, TD learning, which utilizes TD-error, is one of the most effective algorithms for *model-free reinforcement learning*. Therefore, if the CS activities truly contribute to learning the optimal policy or value function through trial-and-error interaction with the environment, it signifies an essential departure from the original MAI model, which assumes *model-based supervised learning* (Chapters 6–8). A partial explanation for the coexistence of the distinct types of CS activities in different studies (Chapters 4,5,6,7,8,9,11, and 12) may be convergent inputs to the IO (Chapter 5) and the current limitation of experimental designs to manipulate these inputs systematically.

Duality of the CS-Induced Plasticity in the Cerebellar Cortex

In previous studies of CS-dependent cerebellar learning, attention has focused on the PF-PC synapse. Consequently, the plasticity of the CNMC has been discussed primarily in terms of CS-dependent LTD at this synapse. However, this viewpoint addresses only part of cerebellar learning.

A



B

Path to PC	(i) Effect on PC	(ii) CS-induced plasticity	(iii) Effect of the plasticity on PC response	(iv) Effect of the plasticity on DCN cell response
a) PF-MLI	inhibition	LTP	↓	↑
b) PF	excitation	LTD	↓	↑

Fig. 11 Effects of CS-induced plasticity on PC and DCN cell. (A) Direct PF input to PC activates PCs (note the plus (+) sign on the synapse), while indirect PF input to PC via MLI inhibits PCs (note the minus (-) sign on the synapse). (B) Concomitant CS activity induces *reciprocal* plasticity on the two pathways: LTD for PF pathway (*LTD/CS*) and LTP for PF-MLI pathway (*LTP/CS*) (i)(ii) [368]. Note that the LTP of PF-MLI inhibition and the LTD of PF excitation provide additive suppression of PC (iii). Finally, the CS-induced PC depression facilitates DCN cell response (= cerebellar output) by *disinhibition* (iv)

It is also known that CS activity induces concurrent and reciprocal long-term potentiation (LTP) in the PF-MLI-PC pathway [368, 369] (Fig. 11). Although the LTP in the PF-MLI-PC pathway has received relatively little attention, the task-related inhibition of PCs by MLIs provides the primary drive to facilitate DCN cells through disinhibition during wrist movements in NHP [357]. Fakharian et al. [253](Chapter 9) also demonstrated that the inhibition of PCs by MLIs excites DCN cells via the exact disinhibition mechanism, halting the saccade at the target. It should also be noted that the PF-MLI-PC pathway has a much lower threshold than the PF-PC pathway [370]. We must devote more effort to studying the physiology of MLIs to understand CS-induced cerebellar learning as suggested before [371].

CS-Induced Effects on DCN Output

Note that the concomitant plasticity induced by CS—LTD at the PF-PC synapse and LTP at the PF-MLI synapse—enhances the activation of DCN cells in response to MF inputs by *reducing* PC inhibition on DCN cells (i.e., disinhibition) (Fig. 11). Overall, while CSs induce LTD in PC responses to MF inputs, this LTD is *reversed* to LTP at the cerebellar outputs (DCN cells) due to the disinhibition of DCN cells. In essence, CS activity enhances concomitant DCN cell outputs by penalizing facilitatory inputs and potentiating inhibitory inputs to PCs. The facilitatory CS effect on the DCN output may resolve the two seemingly contradictory CS activities: one is compatible with model-based supervised learning (i.e., the original MAI model), while the other is compatible with model-free reinforcement learning. Because both are facilitatory on the cerebellar output.

Application of Cerebellar Models To Explain CA

Considering the variability of the input–output organization of CNMC, it is most likely that different types of CNMCs have evolved in different cerebellar regions for different functions. Therefore, they are modeled in different ways (Chapters 4,6,7,8,9,10,11,12,14, and 15).

In Chapter 14, Mitoma and Manto introduced a CNMC model to explain the remarkable capacity of the cerebellum to restore lost function by reorganizing the damaged cerebellar circuitry, i.e., the cerebellar reserve. Specifically, they used a cerebellar Kalman filter model (Chapter 10) for the cerebrocerebellum [334] to consider the function of the cerebellar cortex (prediction) and the DCN (filtering) separately. Although their model is hypothetical and needs to be validated with more clinical and physiological studies, it may provide a unique model-based tool to decode and evaluate CA.

Conclusions and Future Perspectives

The panel of experts agrees that the cerebellum is the *hub* of the central nervous system and is involved in all levels of neural function, ranging from the lowest level of homeostatic functions to the highest level of cognitive-affective functions. The experts also agree that the cerebellum appears to employ a uniform cortical architecture to subserve these diverse functions. This is a unique feature in the neuraxis.

In his famous book "VISION", David Marr (1982) cautioned that "*one has to exercise extreme caution in making inferences from neurophysiological findings about the algorithms and representations being used, particularly until* *one has a clear idea about what information needs to be represented and what processes need to be implemented* (p. 26, lines 11–14 in [372])." Although we have not yet gained a clear understanding of either the representations or the implementations in the cerebellum, we dare to review the expanding frontiers in cerebellar model studies before we proceed toward our ultimate goal: the computational model of the entire cerebellum.

In this Consensus paper, the panel of experts highlighted specific open questions and remaining issues regarding the cerebellar models. We would like to review some of them to suggest our future perspective.

The following specific questions (Q) are among the open questions about the cerebellar models:

(1) Regarding the cerebellar computation in general

Q.Is population coding the correct way to organize the neurons in the cerebellum? If so, what constitutes membership into a population?—The language with which the cerebellum controls behavior appears fundamentally different from the rest of the brain. For example, while neurons in the frontal lobe, parietal lobe, and superior colliculus represent saccades in terms of target location with respect to the fovea, no such representation has been found among the Purkinje cells of the cerebellum. Whereas in the brainstem the premotor neurons such as the burst generators exhibit motor related activity that exhibits strong directional tuning, again no such representation has been found among the Purkinje cells. Thus, it appears that during control of this voluntary movement, the Purkinje cells neither encode behavior in terms of the sensory coordinates, nor in terms of the motor coordinates. In contrast, the Purkinje cells are modulated for all directions and amplitudes of movement, exhibiting only slight changes in the timing of their discharge. Yet, when Purkinje cells are organized into populations, their output appears meaningful, potentially contributing to deceleration and stopping the movement. This implies that individual neurons in the cerebellum are afforded much less autonomy than cells in the rest of the brain in the sense that they must be well organized to play their part in the population.

(2) Regarding reward-related signals in the cerebellum

Q.*Is the cerebellum primarily involved in reinforcement learning?*—There is no consensus on this issue. Thus far, numerous studies have documented cerebellar reward signals and their properties; however, there have been few attempts to make causal manipulations of these signals to test how they influence learning and behavior. It is crucial to identify the causal link from reward-related cerebellar signals (CSs in particular) to the development of reinforced behaviors, as was demonstrated for the CS-dependent adjustment of error in reaching movement [160, see also Chapter 5].

Q. *Where and how are rPEs computed before reaching IO and CFs?*—Recent studies have revealed that CF signals convey prediction errors, including TD errors for sensory events (Chapter 6) and rPEs during reinforcement learning tasks (Chapter 8). In addition to rodent studies, evidence from primate studies also suggests that CF signals encode rewards, expected rewards, and rPEs (Goldberg et al., cited in Chapter 8). However, it is unlikely that rPEs are computed within IO itself. Instead, anatomical evidence suggests that a parvocellular region of RN acts as a key computational hub where cortical information and outputs from the dentate nucleus converge (Chapter 5). Thus, an important open question is how reward and reward prediction signals are assigned to the outputs of the dentate nucleus and cortical areas to generate rPEs within the RN. Elucidating this upstream computational mechanism is critical for fully understanding how cerebellar learning mechanisms might contribute to reinforcement learning processes.

Q.*Could cerebellar signals contribute to rPE computations in the midbrain?*—Given the anatomical connections from the DCN to midbrain dopaminergic structures such as the VTA (Chapters 1 and 8), it is plausible that cerebellar outputs could contribute predictive signals that shape rPE computations in dopaminergic neurons. Investigating whether DCN-derived predictions influence rPE signals in the midbrain will be critical for elucidating the broader role of the cerebellum in reinforcement learning circuits.

Q.*If CFs convey rPEs, do the outputs of DCN represent reward predictions?*—CFs convey reward rPEs, as discussed and agreed upon in the consensus paper. In parallel, the paper emphasizes the cerebellum's general role as a predictor. These considerations together suggest that the outputs of DCN, particularly DN, may represent reward predictions (rP). Although this is a logical inference, direct evidence for rP representations in DCN remains limited. Clarifying whether and how DCN neurons encode predictive reward signals is essential for understanding how the cerebellum contributes to adaptive behavior beyond motor control.

(3) CSs may not be solely for "error coding"

Q.Is there a unified theory that can explain the different response properties of the CSs that would also include the observation that spontaneous CSs' are essential for normal cerebellar function?—PCs demonstrate spontaneous CS activity (~1 Hz) even under anesthesia. Namely, it is not individual CSs that encode an error. Instead, a modulation in CS activities encodes the error. To our knowledge, the functional role of spontaneous CS activities remains unknown.

(4) There remain important but uncharted regions in the cerebellar neuron circuitry

Q. What is the topography and organization of cerebellar connections with the limbic and autonomic systems?

(5) Further studies are needed to clarify the detailed anatomy of these pathways and their functional properties.
(6) We conclude that cerebellar models remain a key-topic of research not only to explain the findings from laboratory and clinical studies, but also to provide novel insights into the understanding of brain functions in physiological and pathological conditions.

Acknowledgements We want to thank Dr. Soichi Nagao for valuable comments on an early draft of Chapter 4.

Authors' contributions Concept and preparation: SK. Draft: all authors. Discussion, corrections and approval of the final version: all authors.

Funding This work was supported by Japan Science and Technology Agency (JST), CREST Grant Number JPMJCR2433 and JSPS KAKENHI Grant Number 25K14517.

Data availability The data shown are available on request.

Declarations

Ethical Approval Not applicable because this is a review article.
Human Ethics and Consent to Participate declarations: not applicable.
Clinical trial number: not applicable.

Competing interests The authors declare no competing interests.

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